

FraudOS — AEGIS.AI

Virtual Humanoid User Flow Specification

AI Execution Specification — Fraud Investigation Happy Flow (MVP) Agent: Rahul

This document is the execution-ready specification for a virtual humanoid (Agent Rahul) operating the FraudOS — AEGIS.AI fraud detection platform. Every screen, element, action, narration, and recovery rule is grounded in the actual MVP UI shown in the attached screenshots and aligned with the Business Requirements Document (V1-MVP-R2). The flow follows the BRD-mandated sequence: real-time monitoring (BR-006) → human-in-the-loop review (BR-004) → daily compliance reporting (BR-005) → model performance review → threshold configuration submission (BR-001 / BR-002).

Field	Value
Application	FraudOS — AEGIS.AI
Agent Name / Code	Rahul
Primary user role	Fraud Investigator (Sarah Chen)
Execution actor	Virtual humanoid operating FraudOS in a browser
Created on	2026-04-29
Flow covered	Fraud Investigation Happy Flow — end-to-end (MVP scope)
Viewport baseline	1456 x 816 browser viewport, 100% zoom
Coordinate origin	Top-left of browser content area (below address bar)
Template status	MVP-scoped, execution-ready
Linked BRD	AI-Powered Financial Anomaly Detection and Reporting (V1-MVP-R2)

1. Prototype Scenario

This document uses FraudOS (running at localhost or production deployment) as the target application. All screens, coordinates, selectors, and narrations are derived from the actual MVP UI and from the BRD UI requirements UI-01 through UI-07.

Area	Detail
Web application	FraudOS — AEGIS.AI
Purpose	AI-powered fraud detection platform with Isolation Forest (primary detection model per BRD Section 1), LSTM (visual indicator only — deferred to Phase 2), Ensemble scoring, Azure OpenAI plain-English explanations (BR-003), human-in-the-loop review (BR-004), threshold submission (BR-001/BR-002), daily compliance reporting (BR-005), and real-time monitoring dashboard (BR-006).
Main desktop URL	http://134.33.132.134/rahul-aegis-fe

Area	Detail
Primary business flow	Investigator opens home (SCR-001-HOME) → reviews dashboard KPIs (SCR-002-DASHBOARD) → inspects flagged transaction queue (SCR-003-TX-QUEUE) → confirms or clears flagged cases → views compliance report (SCR-004-REPORTS) → reviews model performance and audit log (SCR-005-MODEL-PERF) → submits threshold approvals (SCR-006-THRESHOLDS). The new Description page (SCR-007-DESCRIPTION) provides a read-only transaction view available from the taskbar.
Primary audience	A human supervisor watching the humanoid operate FraudOS and explain each decision.
Cultural / language tone	British English. Plain financial language. No ML jargon in investigator-facing output (per BRD Cultural Tone Guidelines).

2. Complete Flow Identification

Field	Required	Description	Filled Value
flow_id	Yes	Unique ID for the business journey.	FLOW-FRAUDOS-INVEST-001
flow_name	Yes	Human-readable name.	FraudOS Fraud Investigation Happy Flow
flow_version	Yes	Version used by AI runtime.	1.0
parent_flow_id	No	Parent flow if this is a child.	None
business_goal	Yes	Why the user wants the flow performed.	Review all flagged transactions, confirm real fraud, clear false positives (BR-004), validate model performance against MET-01 to MET-06, and submit pending threshold changes for approval (BR-001/BR-002).
persona	Yes	Role simulated by humanoid.	Fraud Investigator (Sarah Chen)
trigger_utterance	Yes	User command that starts the flow.	Review today's flagged transactions and explain the fraud findings.
entry_screen_id	Yes	First screen where execution begins.	SCR-001-HOME
expected_end_screen_id	Yes	Final screen after successful execution.	SCR-006-THRESHOLDS
narration_mode	Yes	How much the humanoid speaks.	Detailed guided walkthrough
evidence_required	Yes	Whether screenshots/logs are required.	Yes — capture screenshot at each major screen (per AC-09 audit log completeness)

3. Required Runtime Parameters

Parameter	Required	Type	Example	How AI Uses It
session_id	Yes	String	FRAUDOS-SESSION-001	Correlates all actions, screenshots, and audit log entries (AC-09).
user_role	Yes	String	Fraud Investigator	Determines workspace and explanation level (per BRD Section 4.4 RBAC).
investigator_name	Yes	String	Sarah Chen	Used for audit log validation and Live Activity event matching.
review_date	Yes	String	2026-04-29	Used to verify compliance report date matches today (R-01).
agent_code	Yes	String	AG-MS-0426-004-MVP	Validates dashboard banner reference ID (AG-MS-0426-004 visible top-right of dashboard).
viewport_width	Yes	Number	1456	Used for coordinates and region map.
viewport_height	Yes	Number	816	Used for coordinates and region map.
speaker_style	Yes	Enum	Professional, concise, explanatory	Controls humanoid narration tone (per BRD Tone guideline).
target_fp_rate	Yes	Number	30 (percent)	Compared against current FP Rate KPI (Target <30% per MET-01).
recall_floor	Yes	Number	95 (percent)	Hard floor — used to validate threshold approval safety (MET-04 / Constraint 3).

4. UI/UX Screen Inventory: Where the Humanoid Goes

Each screen ID below maps directly to a discrete screen in the FraudOS application. The same screen IDs are referenced throughout Sections 6 to 20. The humanoid navigates through the seven MVP screens listed below. Notes and Calendar are accessible from the desktop as single-click icons (utility modals, not part of the main investigation flow).

Screen ID	Screen Name	URL / Route	Purpose	Source UI Reference	Expected Visible Evidence
SCR-001-HOME	FraudOS Home Desktop	/	Main app launcher with desktop icons (Notes, Calendar only), taskbar dock with seven module shortcuts, and Live Activity feed (BR-006).	UI Screen 1 (revised)	Two desktop icons: Notes (📝, #desktop-notes-icon) and Calendar (📅, #desktop-calendar-icon); taskbar dock with Desktop, Dashboard, Transactions, Reports, Model Perf, Thresholds, Description; Live Activity panel (#commentaryFeed) with DETECTION / CLEARED events; LIVE badge green; Sarah Chen in top-right.
SCR-002-DASHBOARD	AI Fraud Detection Dashboard	/dashboard	Real-time KPI dashboard satisfying UI-02 / BR-006. Shows banner with agent code, four KPI tiles, flagged transactions chart, risk distribution donut, and flagged transactions table.	UI Screens 2, 3, 4	Banner 'AI Fraud Detection Dashboard — AG-MS-0426-004'; Isolation Forest, LSTM, Azure OpenAI badges; '8 Flagged — 0 Unreviewed' chip; KPI tiles #dashboard-kpi-total, #dashboard-kpi-flagged, #dashboard-kpi-fraud, #dashboard-kpi-fprate showing 8 / 8 / 5 / 37.5%; line chart #dashboard-timeseries-chart over 24 hours; donut #dashboard-riskdist-chart with Critical / High / Medium / Low; flagged table #dashboard-tx-table with 8 transactions.

Screen ID	Screen Name	URL / Route	Purpose	Source UI Reference	Expected Visible Evidence
SCR-003-TX-QUEUE	Flagged Transaction Review Queue	/transactions	Detailed investigator review queue satisfying UI-01 / BR-004. Shows AI explanation column (UI-05), model scores per row, and Fraud / Clear action buttons.	UI Screen 5	Search bar <code>#transactions-search-input</code> ; All Models filter <code>#transactions-model-select</code> ; Any Score filter <code>#transactions-score-select</code> ; All Status filter <code>#transactions-status-select</code> ; transactions table <code>#transactions-table</code> with rows like <code>#tx-row-{id}</code> for TXN-20240424-0064, 0091, 0088; AI Explanation column with critical / contextual narratives; Fraud (<code>#tx-{id}-fraud-btn</code>) and Clear (<code>#tx-{id}-clear-btn</code>) buttons per row.
SCR-004-REPORTS	Compliance Reports	/reports	Daily compliance report and historical archive satisfying UI-06 / BR-005 / R-01.	UI Screen 6	Daily Compliance Report card with: 8 Processed, 8 Flagged, 5 Confirmed Fraud, 37.5% FP Rate, 94.2% Review Completion, 87.0% LLM Helpful Rate; Download PDF button <code>#reports-today-download-btn</code> ; Report Archive table <code>#reports-archive-table</code> with rows <code>#reports-archive-row-{date}</code> for 2026-04-29 through 2026-04-22 with daily metrics and Download buttons <code>#reports-archive-download-btn-{date}</code> .
SCR-005-MODEL-PERF	Model Performance Panel	/model-perf	Model metrics panel satisfying UI-03 / R-02. Shows three model cards, weekly FP rate trend, and full investigator audit log.	UI Screens 7, 8	Three cards: <code>#model-perf-card-isolation_forest</code> (72.0% / 93.0% / 0.813 / 92.0%), <code>#model-perf-card-lstm</code> (91.7% / 95.4% / 0.935 / 24.1%), <code>#model-perf-card-ensemble</code> (93.8% / 97.2% / 0.955 / 21.6%); FP Rate Trend chart <code>#model-fp-chart</code> with Target 30% line; Audit Log <code>#model-audit-table</code> with Sarah Chen entries by TX reference, action, score, timestamp.
SCR-006-THRESHOLDS	Threshold Configuration Panel	/thresholds	Restricted submission interface satisfying UI-04 / BR-001 / BR-002. Shows current vs proposed thresholds, FP / Recall impact, status, and Approve / Revise actions.	UI Screen 9	Admin Role Active badge <code>#threshold-admin-badge</code> ; threshold table <code>#threshold-table</code> with rows <code>#threshold-row-{id}</code> for Crypto Exchange (0.70 → 0.75, Pending), E-commerce High Value (0.78 → 0.78, Approved), High-Value Card Present (0.75 → 0.75, Approved), Small International Transfer (0.80 → 0.86, Pending), Wallet Transfer Domestic (0.82 → 0.82, Blocked); Approve buttons <code>#threshold-approve-btn-{id}</code> ; Rules note <code>#threshold-rules-note</code> referencing Head of Fraud & Risk sign-off and 95% recall floor.
SCR-007-DESCRIPTION	Transaction Description (Read-Only)	/description	Read-only transaction inventory view. Identical column set to the Transactions queue but without action buttons — used for inspection, audit handoff, and stakeholder demos without risk of accidental action.	New screen	WinChrome <code>#description-chrome</code> ; search bar <code>#description-search-input</code> ; filters <code>#description-model-select</code> , <code>#description-score-select</code> , <code>#description-status-select</code> ; description table <code>#description-table</code> with all flagged transaction rows; no Actions column.

5. Screen Layout Region Map

Coordinates are based on a 1456 x 816 viewport. Top-left origin is the browser content area below the address bar. Each region is referenced by Region ID throughout Sections 6 to 9.

Region ID	Applies To (Screen ID)	Pixel Bounds (x1, y1, x2, y2)	Description	Example AI Instruction
REG-TOP-NAV	All screens	0, 0, 1456, 115	Top bar with FraudOS logo, AEGIS.AI label (<code>#menubar</code>), notification bell (<code>#menubar-notification-btn</code>), LIVE badge (<code>#menubar-live-status</code>), user name (<code>#menubar-user-name</code>), avatar (<code>#menubar-avatar</code>), and time.	Verify LIVE badge is green and user shows Sarah Chen before proceeding.

Region ID	Applies To (Screen ID)	Pixel Bounds (x1, y1, x2, y2)	Description	Example AI Instruction
REG-ICON-GRID	SCR-001-HOME	30, 130, 1110, 760	Desktop icon grid (#desktop-grid) — contains only two single-click utility icons: Notes (#desktop-note-s-icon) and Calendar (#desktop-calendar-icon). All other modules moved to the taskbar dock.	Locate Notes or Calendar desktop icon and click once to open the modal. For module navigation use REG-TASKBAR.
REG-LIVE-ACTIVITY	SCR-001-HOME (also visible on SCR-002 to SCR-007)	1135, 130, 1456, 816	Right-rail Live Activity feed (#commentary-section / #commentaryFeed) showing real-time CLEARED and DETECTION events with timestamps. Each event has id #activity-item-{id} .	Read top 3 events to confirm recent activity before opening any screen.
REG-TASKBAR	All screens (persistent dock)	470, 665, 990, 760	Bottom dock (#dock) with seven shortcut icons: Desktop (#dock-desktop), Dashboard (#dock-dashboard), separator, Transactions (#dock-transactions), Reports (#dock-reports), Model Performance (#dock-model), Thresholds (#dock-threshold), Description (#dock-description). This is the primary navigation surface.	Use taskbar to switch modules. The module icons that previously lived on the desktop now live here.
REG-DASH-BANNER	SCR-002-DASHBOARD	30, 200, 1110, 360	Banner with title, agent code (AG-MS-0426-004), Aiven PostgreSQL · Live indicator, model badges, and flagged-count chip.	Read banner to confirm correct screen and verify model badges are present.
REG-KPI-TILES	SCR-002-DASHBOARD	30, 360, 1110, 520	Four KPI cards: Total Transactions in DB, Total Flagged, Confirmed Fraud, False Positive Rate.	Read all four tile values before interpreting dashboard state.
REG-CHART-AREA	SCR-002-DASHBOARD	30, 530, 690, 760	Flagged Transactions Over Time line chart #dashboard-timeseries-chart (last 24 hours · live).	Describe chart trend and peak hours before opening transaction queue.
REG-RISK-DONUT	SCR-002-DASHBOARD	700, 530, 1110, 760	Risk Distribution donut chart #dashboard-riskdist-chart by anomaly score (Critical, High, Medium, Low).	Report dominant risk band and approximate proportion.
REG-TX-TABLE	SCR-002-DASHBOARD, SCR-003-TX-QUEUE, SCR-007-DESCRIPTION	30, 460, 1110, 816	Flagged transactions table with TX ID, Amount, Location, Model, IF Score, LSTM Score, Ensemble, Status, Actions (omitted on SCR-007). Each row carries id #tx-row-{tx_id} .	Read each row top-to-bottom and call out ensemble score and status before any decision.
REG-AI-EXPLAIN	SCR-003-TX-QUEUE, SCR-007-DESCRIPTION	760, 290, 895, 600	AI Explanation column with plain-English fraud reasoning per transaction (BR-003 / Azure OpenAI).	Read AI explanation aloud before clicking Fraud or Clear.
REG-REPORT-METRICS	SCR-004-REPORTS	30, 240, 1110, 460	Daily Compliance Report metrics: Processed, Flagged, Confirmed Fraud, FP Rate, Review Completion, LLM Helpful Rate.	Read all six metrics before opening archive.
REG-REPORT-ARCHIVE	SCR-004-REPORTS	30, 480, 1110, 816	Report Archive table #reports-archive-table with Date, Transactions, Flagged, Fraud, FP Rate, F1 Score, Download.	Verify today's row exists before clicking Download.
REG-MODEL-CARDS	SCR-005-MODEL-PERF	30, 220, 1110, 520	Three model cards (#model-perf-card-isolation_forest , #model-perf-card-lstm , #model-perf-card-ensemble) showing Precision, Recall, F1, FP Rate.	Read all three cards before discussing model performance.
REG-FP-TREND	SCR-005-MODEL-PERF	30, 540, 1110, 850	False Positive Rate Trend Weekly chart #model-fp-chart with Target 30% line (W1 to W8).	Confirm FP Rate trend is descending toward Target line.
REG-AUDIT-LOG	SCR-005-MODEL-PERF	30, 380, 1110, 816	Audit Log table #model-audit-table with Investigator, TX Reference, Action, Score, Timestamp, Notes (AC-09).	Verify Sarah Chen actions are logged with TX references and timestamps.

Region ID	Applies To (Screen ID)	Pixel Bounds (x1, y1, x2, y2)	Description	Example AI Instruction
REG-THRESHOLD-TABLE	SCR-006-THRESHOLDS	70, 280, 1110, 720	Threshold Configuration table #threshold-table with Category, Current, Proposed, FP Impact, Recall Impact, Status, Action.	Read all rows. Validate Admin Role Active badge before clicking Approve.
REG-DESCRIPTION-TABLE	SCR-007-DESCRIPTION	30, 240, 1110, 816	Read-only transactions table #description-table (same data as REG-TX-TABLE without Actions column).	Read rows for inspection only. Do NOT attempt Fraud/Clear actions here — they do not exist on this screen.
REG-MODAL-CENTER	Any modal (Notes #notes-modal , Calendar #calendar-modal , Calculator #calc-modal)	420, 190, 1020, 680	Central modal or pop-up overlay (e.g., session timeout, permission denied, Notes dialog, Calendar dialog).	Read modal title, message, and button labels before clicking.

6. Element Target Registry

The AI prefers stable selectors. Every interactive element carries a unique HTML `id` attribute (added for Playwright automation). These IDs are the **first-choice locators**. Coordinates are fallbacks. Every element references its parent Screen ID. Visibility must be validated before any action.

6A. Button ID and Mapping Registry

This registry lists the buttons and controls that drive flow transitions. Use Button IDs in templates, validations, and recovery steps.

Button ID	Screen ID	Button Label	Button Mapping / Intent	Selector / Locator	Required Action
#dock-dashboard	SCR-001-HOME	Dashboard dock icon	Open AI Fraud Detection Dashboard	#dock-dashboard	Click
#dock-transactions	SCR-001-HOME	Transactions dock icon	Open Flagged Transaction Review Queue	#dock-transactions	Click
#dock-reports	SCR-001-HOME	Reports dock icon	Open Compliance Reports	#dock-reports	Click
#dock-model	SCR-001-HOME	Model Perf dock icon	Open Model Performance Panel	#dock-model	Click
#dock-threshold	SCR-001-HOME	Thresholds dock icon	Open Threshold Configuration Panel	#dock-threshold	Click
#dock-description	SCR-001-HOME	Description dock icon	Open Transaction Description (read-only)	#dock-description	Click
#dock-desktop	SCR-001-HOME	Desktop dock icon	Return to home desktop	#dock-desktop	Click
#desktop-notes-icon	SCR-001-HOME	Notes desktop icon	Open Notes modal	#desktop-notes-icon	Single-click
#desktop-calendar-icon	SCR-001-HOME	Calendar desktop icon	Open Calendar modal	#desktop-calendar-icon	Single-click

Button ID	Screen ID	Button Label	Button Mapping / Intent	Selector / Locator	Required Action
#tx-{tx_id}-fraud-btn	SCR-003-TX-QUEUE	Fraud action button (per row)	Confirm transaction as fraud (BR-004)	#tx-{tx_id}-fraud-btn	Click after reading AI explanation
#tx-{tx_id}-clear-btn	SCR-003-TX-QUEUE	Clear action button (per row)	Clear as false positive (BR-004)	#tx-{tx_id}-clear-btn	Click after reading AI explanation
#tx-{tx_id}-escalate-btn	SCR-003-TX-QUEUE	Escalate action button (per row)	Escalate to supervisor	#tx-{tx_id}-escalate-btn	Click for escalation
#dashboard-pdf-btn	SCR-002-DASHBOARD	PDF export button	Export flagged transaction list as PDF	#dashboard-pdf-btn	Click
#reports-today-download-btn	SCR-004-REPORTS	Download today's PDF button	Download today's compliance report (BR-005)	#reports-today-download-btn	Click
#reports-archive-download-btn-{date}	SCR-004-REPORTS	Archive Download button (per date)	Download archived daily report	#reports-archive-download-btn-{date}	Click
#threshold-approve-btn-{id}	SCR-006-THRESHOLDS	Approve button (per row)	Submit threshold change for sign-off (BR-001 / BR-002)	#threshold-approve-btn-{id}	Click after Admin Role validation
#threshold-revise-btn-{id}	SCR-006-THRESHOLDS	Revise button (per row)	Focus the proposed value input for editing	#threshold-revise-btn-{id}	Click
#{winId}-close-btn	Any modal	Close button (Notes / Calendar / Calculator / WinChrome)	Dismiss modal or page window	#notes-close-btn / #calendar-close-btn / #calc-close-btn / #{winId}-close-btn	Click

6B. Element Target Registry

EL ID	Screen ID	Element Name	Type	Selector / Locator	Fallback Location	Required Action
#desktop-notes-icon	SCR-001-HOME	Notes desktop icon	Icon	#desktop-notes-icon	REG-ICON-GRID x=170 y=260	Single-click to open Notes modal (#notes-modal)
#desktop-calendar-icon	SCR-001-HOME	Calendar desktop icon	Icon	#desktop-calendar-icon	REG-ICON-GRID x=320 y=260	Single-click to open Calendar modal (#calendar-modal)
#dock-dashboard	SCR-001-HOME	Dashboard dock item	Dock Icon	#dock-dashboard	REG-TASKBAR x=560 y=720	Click to open SCR-002-DASHBOARD
#dock-transactions	SCR-001-HOME	Transactions dock item	Dock Icon	#dock-transactions	REG-TASKBAR x=620 y=720	Click to open SCR-003-TX-QUEUE
#dock-reports	SCR-001-HOME	Reports dock item	Dock Icon	#dock-reports	REG-TASKBAR x=680 y=720	Click to open SCR-004-REPORTS
#dock-model	SCR-001-HOME	Model Performance dock item	Dock Icon	#dock-model	REG-TASKBAR x=740 y=720	Click to open SCR-005-MODEL-PERF
#dock-threshold	SCR-001-HOME	Thresholds dock item	Dock Icon	#dock-threshold	REG-TASKBAR x=800 y=720	Click to open SCR-006-THRESHOLDS
#dock-description	SCR-001-HOME	Description dock item	Dock Icon	#dock-description	REG-TASKBAR x=860 y=720	Click to open SCR-007-DESCRIPTION

EL ID	Screen ID	Element Name	Type	Selector / Locator	Fallback Location	Required Action
#commentaryFeed	SCR-001-HOME	Live Activity feed	Panel	#commentaryFeed	REG-LIVE-ACTIVITY	Read top 3 events to confirm real-time data per BR-006
#dashboard-hero-banner	SCR-002-DASHBOARD	Dashboard banner	Banner	#dashboard-hero-banner	REG-DASH-BANNER	Read banner; verify agent code AG-MS-0426-004 and model badges
#dashboard-kpi-total	SCR-002-DASHBOARD	Total Transactions in DB tile	KPI Card	#dashboard-kpi-total	REG-KPI-TILES x=164 y=435	Read value (8); verify it matches DB count
#dashboard-kpi-flagged	SCR-002-DASHBOARD	Total Flagged tile	KPI Card	#dashboard-kpi-flagged	REG-KPI-TILES x=430 y=435	Read value (8); confirm ensemble score >0.80 threshold per UI-02
#dashboard-kpi-fraud	SCR-002-DASHBOARD	Confirmed Fraud tile	KPI Card	#dashboard-kpi-fraud	REG-KPI-TILES x=695 y=435	Read confirmed fraud count (5) — investigator confirmed (BR-004)
#dashboard-kpi-fprate	SCR-002-DASHBOARD	False Positive Rate tile	KPI Card	#dashboard-kpi-fprate	REG-KPI-TILES x=890 y=435	Read FP Rate (37.5%); compare to Target <30% (MET-01)
#dashboard-timeseries-chart	SCR-002-DASHBOARD	Time-series chart	Chart	#dashboard-timeseries-chart	REG-CHART-AREA	Describe peak hours and flagged-vs-confirmed gap
#dashboard-riskdist-chart	SCR-002-DASHBOARD	Risk distribution donut chart	Chart	#dashboard-riskdist-chart	REG-RISK-DONUT	Name dominant risk band
#dashboard-tx-table	SCR-002-DASHBOARD	Dashboard transactions table	Table	#dashboard-tx-table	REG-TX-TABLE	Read each row; cite TXN-20240424-0064 (\$34,200 / Hong Kong / Ens 0.97)
#dashboard-search-input	SCR-002-DASHBOARD	Dashboard search input	Input	#dashboard-search-input	REG-TX-TABLE x=240 y=478	Type TX ID or location to filter
#dashboard-status-select	SCR-002-DASHBOARD	Dashboard status filter	Dropdown	#dashboard-status-select	REG-TX-TABLE x=920 y=478	Filter by status
#dashboard-pdf-btn	SCR-002-DASHBOARD	PDF export button	Button	#dashboard-pdf-btn	REG-TX-TABLE x=1040 y=478	Click to export flagged transaction list as PDF
#transactions-search-input	SCR-003-TX-QUEUE	Search bar	Input	#transactions-search-input	REG-TOP-NAV x=390 y=228	Type TX ID or location to filter queue
#transactions-model-select	SCR-003-TX-QUEUE	Model filter	Dropdown	#transactions-model-select	REG-TOP-NAV x=762 y=228	Filter queue by Isolation Forest, LSTM, or Ensemble
#transactions-score-select	SCR-003-TX-QUEUE	Score filter	Dropdown	#transactions-score-select	REG-TOP-NAV x=920 y=228	Filter by High / Medium / Low score band
#transactions-status-select	SCR-003-TX-QUEUE	Status filter	Dropdown	#transactions-status-select	REG-TOP-NAV x=1080 y=228	Filter by Unreviewed / Fraud / Cleared / Escalated

EL ID	Screen ID	Element Name	Type	Selector / Locator	Fallback Location	Required Action
#transactions-table	SCR-003-TX-QUEUE	Transactions table	Table	#transactions-table	REG-TX-TABLE	Read merchant, device, AI explanation, scores per row
#tx-row-{tx_id}	SCR-003-TX-QUEUE	Transaction row (per tx)	Table Row	#tx-row-{tx_id} (e.g., #tx-row-1)	REG-TX-TABLE	Locate row by TX id before action
#tx-{tx_id}-fraud-btn	SCR-003-TX-QUEUE	Fraud action button (per row)	Button	#tx-{tx_id}-fraud-btn	REG-TX-TABLE x=999 y={row_y}	Click after reading AI explanation to confirm fraud (BR-004)
#tx-{tx_id}-clear-btn	SCR-003-TX-QUEUE	Clear action button (per row)	Button	#tx-{tx_id}-clear-btn	REG-TX-TABLE x=1047 y={row_y}	Click after reading AI explanation to clear as false positive (BR-004)
#tx-{tx_id}-escalate-btn	SCR-003-TX-QUEUE	Escalate action button (per row)	Button	#tx-{tx_id}-escalate-btn	REG-TX-TABLE x=1095 y={row_y}	Click to escalate to supervisor
#reports-today-download-btn	SCR-004-REPORTS	Today's Download PDF button	Button	#reports-today-download-btn	REG-REPORT-METRICS x=1004 y=234	Click to download today's compliance report (R-01)
#reports-archive-table	SCR-004-REPORTS	Archive table	Table	#reports-archive-table	REG-REPORT-ARCHIVE	Verify today's row exists
#reports-archive-row-{date}	SCR-004-REPORTS	Archive row (per date)	Table Row	#reports-archive-row-{date} (e.g., #reports-archive-row-2026-04-29)	REG-REPORT-ARCHIVE	Locate the date row
#reports-archive-download-btn-{date}	SCR-004-REPORTS	Archive Download button (per date)	Button	#reports-archive-download-btn-{date}	REG-REPORT-ARCHIVE x=1010 y={row_y}	Click to download an archived daily report
#model-perf-card-isolation_forest	SCR-005-MODEL-PERF	Isolation Forest model card	Card	#model-perf-card-isolation_forest	REG-MODEL-CARDS x=170 y=290	Read Precision, Recall, F1, FP Rate (R-02)
#model-perf-card-lstm	SCR-005-MODEL-PERF	LSTM model card	Card	#model-perf-card-lstm	REG-MODEL-CARDS x=540 y=290	Read Precision, Recall, F1, FP Rate (R-02)
#model-perf-card-ensemble	SCR-005-MODEL-PERF	Ensemble model card	Card	#model-perf-card-ensemble	REG-MODEL-CARDS x=915 y=290	Read Precision, Recall, F1, FP Rate (R-02)
#model-fp-chart	SCR-005-MODEL-PERF	FP Rate trend chart	Chart	#model-fp-chart	REG-FP-TREND	Confirm trend descending toward Target 30% line
#model-audit-table	SCR-005-MODEL-PERF	Audit Log table	Table	#model-audit-table	REG-AUDIT-LOG	Verify Sarah Chen entries with TX references and timestamps (AC-09)
#model-audit-row-{id}	SCR-005-MODEL-PERF	Audit Log row (per id)	Table Row	#model-audit-row-{id}	REG-AUDIT-LOG	Read individual decision entry
#threshold-admin-badge	SCR-006-THRESHOLDS	Admin Role Active badge	Badge	#threshold-admin-badge	REG-THRESHOLD-TABLE x=1330 y=290	Confirm badge is visible before clicking any Approve button (UI-04)

EL ID	Screen ID	Element Name	Type	Selector / Locator	Fallback Location	Required Action
#threshold-table	SCR-006-THRESHOLDS	Threshold table	Table	#threshold-table	REG-THRESHOLD-TABLE	Read all category rows
#threshold-row-{id}	SCR-006-THRESHOLDS	Threshold row (per id)	Table Row	#threshold-row-{id}	REG-THRESHOLD-TABLE	Locate the row by category id
#threshold-input-{id}	SCR-006-THRESHOLDS	Threshold proposed-value input (per id)	Input	#threshold-input-{id}	REG-THRESHOLD-TABLE x=620 y={row_y}	Type or focus to revise the proposed threshold value (0.50–0.99)
#threshold-approve-btn-{id}	SCR-006-THRESHOLDS	Approve button (per id)	Button	#threshold-approve-btn-{id}	REG-THRESHOLD-TABLE x=950 y={row_y}	Click to submit threshold change request (BR-001 / BR-002)
#threshold-revise-btn-{id}	SCR-006-THRESHOLDS	Revise button (per id)	Button	#threshold-revise-btn-{id}	REG-THRESHOLD-TABLE x=950 y={row_y}	Click to focus the input for revision
#threshold-row-{id}.t-status.blocked	SCR-006-THRESHOLDS	Wallet Transfer blocked status	Status Badge	#threshold-row-{id}.t-status.blocked	REG-THRESHOLD-TABLE x=1080 y=665	Validate-only — do not click. Recall risk auto-block per Constraint 3.
#threshold-rules-note	SCR-006-THRESHOLDS	Rules note	Text Block	#threshold-rules-note	REG-THRESHOLD-TABLE bottom	Read aloud once: '95% recall floor, Head of Fraud & Risk sign-off required'
#description-search-input	SCR-007-DESCRIPTION	Description search input	Input	#description-search-input	REG-DESCRIPTION-TABLE x=240 y=180	Type TX ID or location to filter
#description-model-select	SCR-007-DESCRIPTION	Description model filter	Dropdown	#description-model-select	REG-DESCRIPTION-TABLE x=620 y=180	Filter by model
#description-score-select	SCR-007-DESCRIPTION	Description score filter	Dropdown	#description-score-select	REG-DESCRIPTION-TABLE x=780 y=180	Filter by score band
#description-status-select	SCR-007-DESCRIPTION	Description status filter	Dropdown	#description-status-select	REG-DESCRIPTION-TABLE x=940 y=180	Filter by status
#description-table	SCR-007-DESCRIPTION	Description table	Table	#description-table	REG-DESCRIPTION-TABLE	Read rows for inspection only — no Actions column present
#notes-modal	Any	Notes modal	Modal	#notes-modal	REG-MODAL-CENTER	Opened from #desktop-notes-icon ; close via #notes-close-btn
#calendar-modal	Any	Calendar modal	Modal	#calendar-modal	REG-MODAL-CENTER	Opened from #desktop-calendar-icon ; close via #calendar-close-btn
#{winId}-close-btn	Any	Page close button (WinChrome)	Button	#{winId}-close-btn (e.g., #dashboard-close-btn, #description-close-btn)	Top-left of WinChrome	Click to return to SCR-001-HOME

7. Standard AI Action Record Template

Every action in Sections 8 and 9 follows this JSON structure. The template extends the master template with `button_id` and `button_mapping` so that button-driven steps can be referenced from the Button ID Registry (Section 6A).

```
{
  "action_id": "ACT-###",
  "screen_id": "SCR-###",
  "action_type": "click | type | select | scroll | explain_screen | validate | recover",
  "target": {
    "element_id": "EL-###",
    "button_id": "BTN-###",
    "selector": "preferred stable locator (HTML id first, then aria-label, then text)",
    "button_mapping": "registry mapping or intent label",
    "fallback_region": "REG-...",
    "fallback_coordinates": {"x": 0, "y": 0}
  },
  "input_parameters": {},
  "preconditions": [],
  "execution_steps": [],
  "expected_response": {},
  "validation_rules": [],
  "humanoid_speaker_notes": {
    "before_action": "",
    "during_action": "",
    "after_action": ""
  },
  "failure_recovery": []
}
```

8. Detailed Examples for All Action Types

Each example below demonstrates one action type using a real screen, real element, and real data from the FraudOS UI. Action IDs in this section are illustrative; the canonical execution sequence is in Section 9.

8A. Navigation — Open Dashboard from Home Screen via Taskbar Dock (SCR-001-HOME → SCR-002-DASHBOARD)

Field	Value
action_id	ACT-001
screen_id	SCR-001-HOME
action_type	click
target element	EL-003 — Dashboard dock item

Field	Value
button_id	#dock-dashboard
selector	#dock-dashboard
fallback region / coordinates	REG-TASKBAR x=560 y=720
preconditions	FraudOS home screen loaded; LIVE badge is green; Sarah Chen visible top-right; Live Activity feed shows at least one event; taskbar dock is visible at the bottom of the viewport.
execution steps	1. Confirm LIVE badge and user name in REG-TOP-NAV. 2. Locate the Dashboard icon in the taskbar dock (#dock-dashboard). 3. Click the icon.
expected response	Dashboard screen (SCR-002-DASHBOARD) loads; banner reads 'AI Fraud Detection Dashboard — AG-MS-0426-004'; four KPI tiles visible (#dashboard-kpi-total , #dashboard-kpi-flagged , #dashbo rd-kpi-fraud , #dashboard-kpi-fprate).
validation rules	Banner contains 'AI Fraud Detection Dashboard'; agent code AG-MS-0426-004 is visible top-right; Isolation Forest, LSTM, Azure OpenAI badges all present; '8 Flagged — 0 Unreviewed' chip visible.
before_action narration	I am on the FraudOS home screen. The desktop now shows only Notes and Calendar utility icons. I will use the taskbar dock at the bottom to open the Dashboard.
after_action narration	The AI Fraud Detection Dashboard is now open. I can see the agent code AG-MS-0426-004, model badges for Isolation Forest, LSTM, and Azure OpenAI, and four KPI tiles for transactions, flagged, confirmed fraud, and false positive rate.
failure_recovery	If #dock-dashboard is not present, refresh the page once. If still missing, navigate directly to /dashboard via the URL bar. If the dashboard does not load after one refresh, report the screen as unreachable and stop the flow (ERR-001).

8B. Explain Screen — Read KPI Tiles on Dashboard (SCR-002-DASHBOARD)

Field	Value
action_id	ACT-002
screen_id	SCR-002-DASHBOARD
action_type	explain_screen
target region	REG-KPI-TILES (EL-011, EL-012, EL-013, EL-014)
selector	#dashboard-kpi-total , #dashboard-kpi-flagged , #dashboard-kpi-fraud , #dashboard-kpi-fprate
preconditions	Dashboard loaded. All four KPI tiles visible with numeric values.
execution steps	1. Read EL-011 'Total Transactions in DB' (#dashboard-kpi-total). 2. Read EL-012 'Total Flagged' (#dashboard-kpi-flagged). 3. Read EL-013 'Confirmed Fraud' (#dashboard-kpi-fraud). 4. Read EL-014 'False Positive Rate' (#dashboard-kpi-fprate) and compare to 'Target <30%' label.
validation rules	All four tiles have non-empty numeric values. FP Rate tile shows 'Target <30%' label. KPI values are consistent with Live Activity panel events.
narration	Looking at today's KPIs: there are 8 total transactions in the database, all 8 have been flagged by the ensemble model with score above 0.80, 5 have been confirmed as fraud by the investigator, and the false positive rate is currently 37.5 percent. The target is below 30 percent, so the system is above target. This tells us the ensemble is catching fraud but is also flagging some legitimate transactions, which we will need to address through threshold tuning.

Field	Value
failure_recovery	If any tile shows no value or a loading spinner, wait 10 seconds and refresh the page once. If values remain empty after refresh, report the tile as unavailable and continue with visible tiles. Do not infer missing values.

8C. Click Action — Confirm Fraud Decision on Transaction Row (SCR-003-TX-QUEUE)

Field	Value
action_id	ACT-003
screen_id	SCR-003-TX-QUEUE
action_type	click + validate
target element	EL-027 — Fraud button for TXN-20240424-0064
button_id	<code>#tx-{tx_id}-fraud-btn</code>
selector	<code>#tx-{id_of_TXN-20240424-0064}-fraud-btn</code> (resolve numeric <code>tx.id</code> from the row before clicking)
fallback region / coordinates	REG-TX-TABLE x=999 y=371
preconditions	Transaction row visible (<code>#tx-row-{id}</code>). AI explanation read (REG-AI-EXPLAIN). Ensemble score 0.97 (Critical band). Current row status shows 'Fraud' pending investigator confirmation.
execution steps	1. Read AI explanation: 'CRITICAL: Wire transfer of \$34,200 initiated from a new device in Hong Kong.' 2. Verify ensemble score 0.97. 3. Click Fraud button (<code>#tx-{id}-fraud-btn</code>). 4. Validate row status updates. 5. Verify Live Activity panel adds new DETECTION event 'Sarah Chen confirmed fraud: 0064 (\$34,200.00)' as a new <code>#activity-item-{id}</code> entry.
validation rules	Row status shows 'Fraud' (confirmed). Live Activity shows new DETECTION entry within 1 second (per AC-06). Confirmed Fraud KPI tile (<code>#dashboard-kpi-fraud</code>) increments by 1 on the dashboard. Audit log (<code>#model-audit-table</code>) on SCR-005-MODEL-PERF records action with investigator ID, decision, and timestamp (AC-09).
before_action narration	I am reviewing transaction TXN-20240424-0064. The AI explanation says this is a critical-risk wire transfer of \$34,200 initiated from a new device in Hong Kong. The ensemble score is 0.97, the highest confidence band. I will confirm this as fraud per BR-004 human-in-the-loop review.
after_action narration	Fraud confirmed. The status has been updated and the Live Activity panel on the right shows this action was recorded. The Confirmed Fraud count on the dashboard will now reflect this decision.
failure_recovery	If the Fraud button is unresponsive, scroll to ensure the row is fully in the viewport and retry once. If status does not update within 5 seconds, refresh the page and check the Audit Log on SCR-005-MODEL-PERF for the action (ERR-003). If audit shows no entry, escalate to supervisor.

8D. Click Action — Mark False Positive on Transaction Row (SCR-003-TX-QUEUE)

Field	Value
action_id	ACT-004
screen_id	SCR-003-TX-QUEUE
action_type	click + validate
target element	EL-028 — Clear button for TXN-20240424-0091

Field	Value
button_id	#tx-{tx_id}-clear-btn
selector	#tx-{id_of_TXN-20240424-0091}-clear-btn
fallback region / coordinates	REG-TX-TABLE x=1047 y=423
preconditions	Transaction row visible. AI explanation read: 'Amount 6x usual spend in new country with unfamiliar device. Geolocation does not match profile.' Ensemble score 0.92. Status column shows 'Cleared' from prior investigator review context.
execution steps	1. Read AI explanation for TXN-20240424-0091. 2. Note ensemble score 0.92. 3. Confirm context supports legitimate transaction. 4. Click Clear button (#tx-{id}-clear-btn). 5. Validate Live Activity shows CLEARED event 'Sarah Chen cleared as false positive: 0091'.
validation rules	Row status confirmed as 'Cleared'. Live Activity logs CLEARED event. Decision recorded in audit log (AC-09). Action recorded for retraining feedback loop (BR-004).
before_action narration	Transaction TXN-20240424-0091 is an \$8,400 purchase at an electronics store in Lagos, Nigeria. Although the amount is high and the location is new, the investigator context confirms this is a legitimate transaction. I will clear this as a false positive.
after_action narration	Cleared. The false positive has been recorded. This decision feeds into the retraining pipeline as a negative example per BR-004 and contributes to the 37.5 percent false positive rate visible on the dashboard.
failure_recovery	If the Clear button is greyed out, check whether a Fraud decision was already recorded for this row. If so, do not override without supervisor confirmation. If status does not change within 5 seconds, retry once (ERR-003).

8E. Select — Apply Filter on Transaction Queue (SCR-003-TX-QUEUE)

Field	Value
action_id	ACT-005
screen_id	SCR-003-TX-QUEUE
action_type	select
target element	EL-022 — Model filter dropdown
selector	#transactions-model-select
fallback region / coordinates	REG-TOP-NAV x=762 y=228
execution steps	1. Click the Model filter dropdown (#transactions-model-select). 2. Select 'Ensemble' to filter by ensemble-scored transactions. 3. Confirm #transactions-table refreshes to show only Ens-labeled rows.
expected response	Transaction table refreshes; only rows with Ens model label are shown.
before_action narration	I will filter the queue to show only Ensemble model detections, since these combine Isolation Forest and LSTM signals and have the highest confidence.
after_action narration	The queue now shows only Ensemble-scored transactions. These are the highest-confidence fraud signals for review.

Field	Value
failure_recovery	If the dropdown does not open, click another filter (<code>#transactions-status-select</code> or <code>#transactions-score-select</code>) to confirm the UI is responsive. If the filter does not change the table, clear the filter and proceed with the unfiltered queue.

8F. Validate — Threshold Approval Pre-Check (SCR-006-THRESHOLDS)

Field	Value
action_id	ACT-006
screen_id	SCR-006-THRESHOLDS
action_type	validate
target element	EL-040 — Admin Role Active badge
selector	<code>#threshold-admin-badge</code>
fallback region / coordinates	REG-THRESHOLD-TABLE x=1330 y=290
preconditions	Threshold Configuration Panel loaded (SCR-006-THRESHOLDS). Five threshold rows visible in <code>#threshold-table</code> .
execution steps	1. Verify <code>#threshold-admin-badge</code> is visible. 2. For each <code>#threshold-row-{id}</code> , read Current, Proposed, FP Impact, Recall Impact, Status. 3. Identify rows with Status = 'Pending'. 4. For each Pending row, validate Recall Impact is within tolerance (must not reduce recall below 95% per MET-04 / Constraint 3).
validation rules	Admin Role Active badge present; Pending rows have non-blocking Recall Impact (Crypto Exchange -0.8% and Small International -1.2% are within tolerance); Wallet Transfer Domestic Status correctly shows 'Blocked' due to -3.1% recall risk.
narration	I am on the Threshold Configuration Panel. The Admin Role Active badge is visible. I can see five categories: Crypto Exchange and Small International Transfer are pending approval; E-commerce High Value and High-Value Card Present are already approved; and Wallet Transfer Domestic is automatically blocked because the proposed change would reduce recall by 3.1 percent, exceeding the 95 percent recall floor defined in MET-04.
failure_recovery	If <code>#threshold-admin-badge</code> is absent, do not click any Approve button. Report role limitation to supervisor and stop (DEC-004 / ERR-005). If Status column is unreadable, refresh page once.

8G. Click Action — Open Read-Only Description Page (SCR-001-HOME → SCR-007-DESCRIPTION)

Field	Value
action_id	ACT-007
screen_id	SCR-001-HOME
action_type	click
target element	EL-008 — Description dock item
button_id	<code>#dock-description</code>
selector	<code>#dock-description</code>

Field	Value
fallback region / coordinates	REG-TASKBAR x=860 y=720
preconditions	Home screen visible; taskbar dock visible.
execution steps	1. Locate <code>#dock-description</code> in the taskbar. 2. Click the icon. 3. Confirm SCR-007-DESCRIPTION opens with <code>#description-table</code> populated.
expected response	Transaction Description window opens; <code>#description-table</code> shows rows; no Actions column present.
before_action narration	I will open the Description page from the taskbar to review the transaction inventory in read-only mode without risk of accidental action.
after_action narration	The Description page is open. I can see the same transaction columns as the Transactions queue, but the Actions column is intentionally absent because this is a read-only view.
failure_recovery	If <code>#dock-description</code> is missing, refresh the page once. If still missing, navigate to <code>/description</code> directly. If the table is empty, clear all filters in <code>#description-search-input</code> , <code>#description-model-select</code> , <code>#description-score-select</code> , <code>#description-status-select</code> and retry.

9. Complete Happy Flow Action Matrix

This matrix defines every step of the Fraud Investigation Happy Flow in execution order. Each row references its Screen ID, Element ID, Button ID (where applicable), and BRD reference. The flow follows: SCR-001 → SCR-002 → SCR-003 → SCR-004 → SCR-005 → SCR-006. SCR-007-DESCRIPTION is an optional inspection step (sequence 12a) that may be inserted between the queue review and the compliance report.

Seq	Action ID	Screen ID	Action Type	Target EL	Button ID	Selector / Location	Expected Response	Speaker Notes (summary)	Fallback
1	ACT-001	SCR-001-HOME	explain_screen	<code>#commentaryFeed</code>	—	<code>#commentaryFeed</code> (REG-LIVE-ACTIVITY)	Live Activity feed visible with DETECTION / CLEARED events	I can see the FraudOS home with two desktop icons (Notes, Calendar) and recent fraud events in the Live Activity feed.	Reload page if LIVE badge is red
2	ACT-002	SCR-001-HOME	click	<code>#dock-dashboard</code>	<code>#dock-dashboard</code>	<code>#dock-dashboard</code>	Dashboard (SCR-002-DASHBOARD) loads with banner and KPI tiles	Opening the Dashboard from the taskbar to see today's fraud metrics.	Refresh page; navigate to <code>/dashboard</code> directly if dock missing
3	ACT-003	SCR-002-DASHBOARD	explain_screen	EL-011 to EL-014	—	<code>#dashboard-kpi-total</code> , <code>#dashboard-kpi-flagged</code> , <code>#dashboard-kpi-fraud</code> , <code>#dashboard-kpi-fprate</code>	All four KPI values read and narrated	8 transactions, 8 flagged, 5 confirmed fraud, 37.5% FP Rate (above 30% target per MET-01).	If tile blank: wait 10s and refresh
4	ACT-004	SCR-002-DASHBOARD	explain_screen	EL-015, EL-016	—	<code>#dashboard-timeseries-chart</code> + <code>#dashboard-riskdist-chart</code>	Chart trend described; donut color distribution narrated	Chart peaks at 14:00–18:00. Risk donut is predominantly High (orange) and Critical (red).	Zoom browser if chart labels are too small

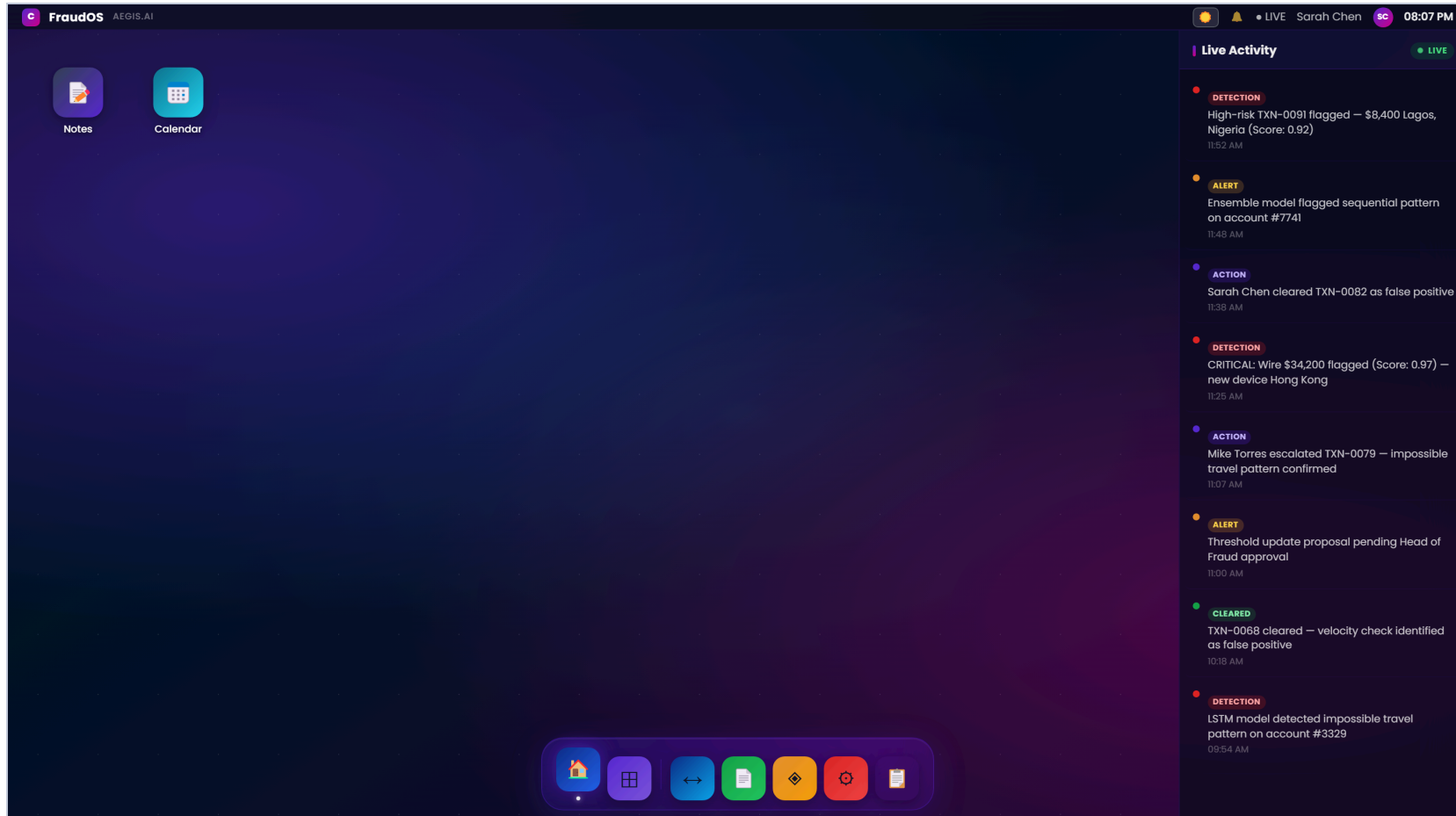
Seq	Action ID	Screen ID	Action Type	Target EL	Button ID	Selector / Location	Expected Response	Speaker Notes (summary)	Fallback
5	ACT-005	SCR-002-DASHBOARD	explain_screen	#dashboard-tx-table	—	#dashboard-tx-table (REG-TX-TABLE)	All 8 transaction rows read with scores and statuses	Reviewing flagged table: TXN-0064 at \$34,200 Hong Kong has ensemble 0.97 (highest). TXN-0079 at \$15,800 Dubai is cleared.	Scroll table if rows extend below viewport
6	ACT-006	SCR-001-HOME	click	#dock-transactions	#dock-transactions	#dock-transactions	Transaction Review Queue (SCR-003-TX-QUEUE) opens with full detail table	Opening the detailed review queue per UI-01 / BR-004 from the taskbar dock.	Refresh page; navigate to /transactions directly
7	ACT-007	SCR-003-TX-QUEUE	explain_screen	#transactions-table	—	#transactions-table + REG-AI-EXPLAIN	AI explanation column visible per row	Queue shows merchant, device, AI explanation, and model scores per transaction.	Scroll right if AI explanation column is hidden
8	ACT-008	SCR-003-TX-QUEUE	click	#tx-{tx_id}-fraud-btn	#tx-{tx_id}-fraud-btn	#tx-{id_of_TXN-0064}-fraud-btn	Status updates; Live Activity logs DETECTION event (AC-06)	Confirming TXN-0064 Hong Kong wire transfer as fraud — ensemble 0.97, Critical.	Retry once if button unresponsive (ERR-003)
9	ACT-009	SCR-003-TX-QUEUE	click	#tx-{tx_id}-clear-btn	#tx-{tx_id}-clear-btn	#tx-{id_of_TXN-0091}-clear-btn	Status updates to Cleared; Live Activity logs CLEARED event	Clearing TXN-0091 Lagos electronics — investigator determined this is legitimate.	Check if already cleared before clicking
10	ACT-010	SCR-003-TX-QUEUE	click	#tx-{tx_id}-clear-btn	#tx-{tx_id}-clear-btn	#tx-{id_of_TXN-0088}-clear-btn	Status updates to Cleared; Live Activity logs event	Clearing TXN-0088 Moscow crypto exchange — card-testing pattern resolved by investigator.	Verify AI explanation before clearing
11	ACT-011	SCR-001-HOME	click	#dock-reports	#dock-reports	#dock-reports	Compliance Reports (SCR-004-REPORTS) opens	Opening Compliance Reports from the taskbar to verify today's metrics per UI-06 / R-01.	Navigate to /reports directly if dock missing
12	ACT-012	SCR-004-REPORTS	explain_screen	#reports-archive-table	—	REG-REPORT-METRICS + #reports-archive-table	All six daily metrics narrated; archive rows visible	Today (2026-04-29): 8 processed, 8 flagged, 3 fraud (live), 62.5% FP Rate at this point. Yesterday (2026-04-28): 8 / 8 / 5 / 37.5%.	If metrics blank: refresh page once
12a	ACT-012a	SCR-001-HOME → SCR-007-DESCRIPTION	click (optional)	#dock-description	#dock-description	#dock-description	Description page opens read-only	Optional read-only inspection: opening Description to share inventory with stakeholder without action risk.	Skip if not requested
13	ACT-013	SCR-004-REPORTS	click	#reports-today-download-btn	#reports-today-download-btn	#reports-today-download-btn	Today's compliance report PDF downloads	Downloading today's compliance report per BR-005.	If download fails: alert compliance-alerts@microsoft.com per Guardrail 6

Seq	Action ID	Screen ID	Action Type	Target EL	Button ID	Selector / Location	Expected Response	Speaker Notes (summary)	Fallback
14	ACT-014	SCR-001-HOME	click	#dock-model	#dock-model	#dock-model	Model Performance Panel (SCR-005-MODEL-PERF) opens	Opening Model Performance from the taskbar to review metrics per UI-03 / R-02.	Navigate to /model-perf directly if dock missing
15	ACT-015	SCR-005-MODEL-PERF	explain_screen	EL-034, EL-035, EL-036, EL-037	—	#model-perf-card-isolation_forest, #model-perf-card-lstm, #model-perf-card-ensemble, #model-fp-chart	All three model cards read; FP trend chart described	Ensemble leads: 93.8% precision, 97.2% recall, F1 0.955, FP Rate 21.6%. FP Rate Trend descending from W1 (~85%) toward 30% Target line at W8.	Zoom if F1 score values are too small
16	ACT-016	SCR-005-MODEL-PERF	explain_screen	#model-audit-table	—	#model-audit-table	Investigator decision log visible with timestamps and actions	Audit log confirms all decisions: TXN-0064 FRAUD 0.97 at 12:32:24, TXN-0079 CLEARED 0.85 at 12:33:13, etc. (per AC-09).	Scroll down if log rows exceed viewport
17	ACT-017	SCR-001-HOME	click	#dock-threshold	#dock-threshold	#dock-threshold	Threshold Configuration Panel (SCR-006-THRESHOLDS) opens	Checking threshold configuration from the taskbar per UI-04 / BR-001.	Navigate to /thresholds directly if dock missing
18	ACT-018	SCR-006-THRESHOLDS	validate	#threshold-admin-badge	—	#threshold-admin-badge	Admin badge visible; rows readable	Admin Role is active. Reviewing five threshold categories before approving.	If badge absent: stop, do not approve (DEC-004)
19	ACT-019	SCR-006-THRESHOLDS	click	#threshold-approve-btn-{id}	#threshold-approve-btn-{id}	#threshold-approve-btn-{id_of_crypto_exchange}	Status changes from Pending to Approved (submitted to BR-002 workflow)	Approving Crypto Exchange threshold submission 0.70 → 0.75. FP impact -10%, recall impact -0.8% (within 95% floor).	Do not approve if recall impact ≤ -5%
20	ACT-020	SCR-006-THRESHOLDS	click	#threshold-approve-btn-{id}	#threshold-approve-btn-{id}	#threshold-approve-btn-{id_of_small_international}	Status changes from Pending to Approved	Approving Small International Transfer 0.80 → 0.86. FP -22%, recall -1.2% (within tolerance).	Escalate to Head of Fraud & Risk if recall ≤ -3%
21	ACT-021	SCR-006-THRESHOLDS	validate	#threshold-row-{id}.t-status.blocked	—	#threshold-row-{id_of_wallet_transfer}.t-status.blocked	Status remains Blocked; Action shows 'Recall risk — blocked'	Wallet Transfer Domestic correctly blocked by system: -3.1% recall risk exceeds 95% recall floor (Constraint 3).	Do not attempt to override blocked status

10. UI/UX Screen Visual Inventory

Each FraudOS UI screen below is referenced by Screen ID throughout this document. Live screenshots from the running frontend are the authoritative source for all coordinates and visual content. Screenshots captured at production URL <http://134.33.132.134/rahul-aegis-fe>.

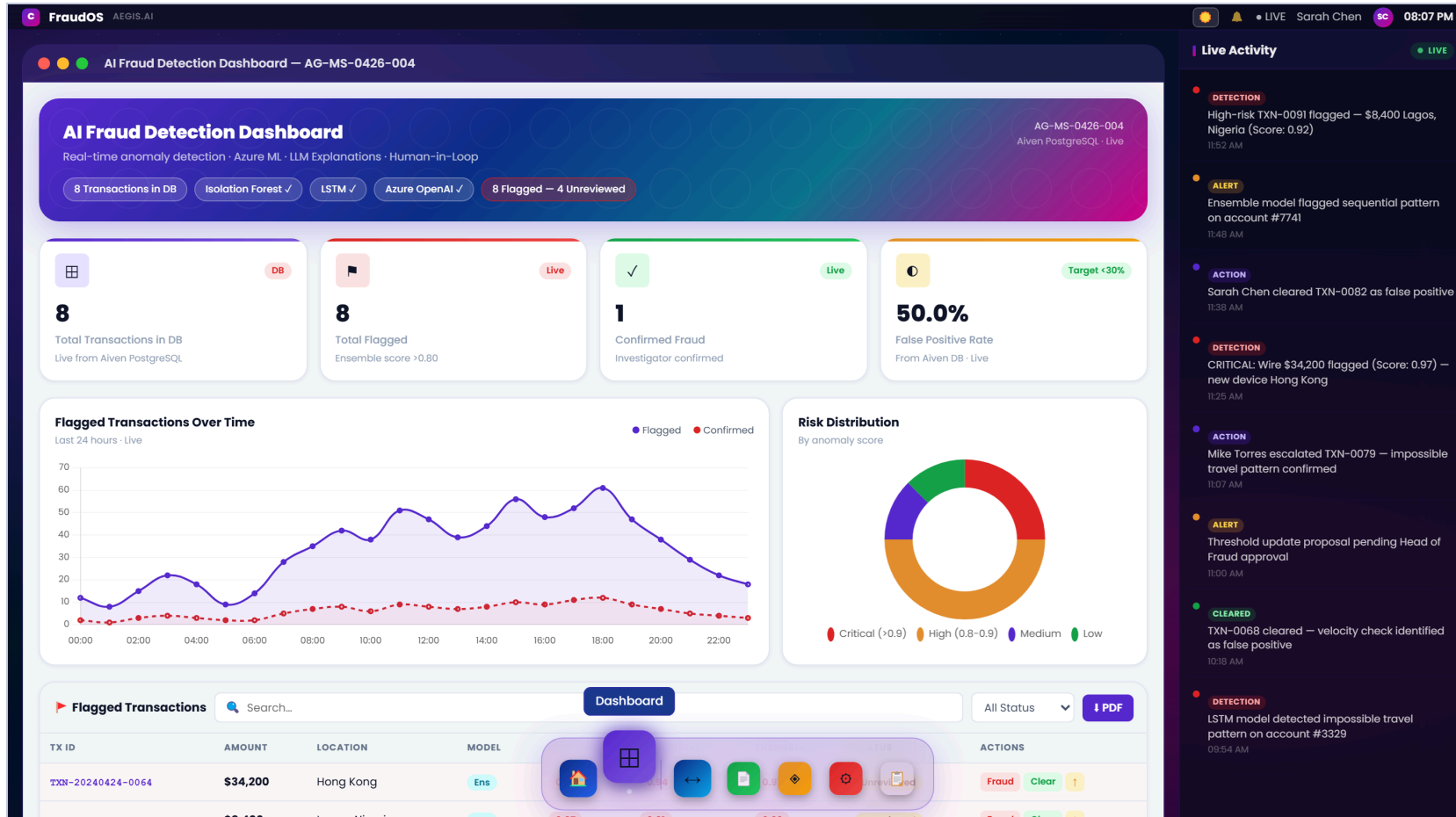
10.1 SCR-001-HOME — FraudOS Home Desktop



Layout (left to right, top to bottom):

- Top bar** (REG-TOP-NAV, `#menubar`): FraudOS logo (C), AEGIS.AI label, theme toggle, notification bell (`#menubar-notification-btn`), green LIVE indicator (`#menubar-live-status`), user name "Sarah Chen", avatar (SC), clock "07:51 PM".
- Desktop icon grid** (REG-ICON-GRID, `#desktop-grid`): only two utility icons visible — Notes (`#desktop-notes-icon`) and Calendar (`#desktop-calendar-icon`). All module navigation has moved to the dock.
- Taskbar dock** (REG-TASKBAR, `#dock`) at the bottom centre, containing seven items in order: Desktop (`#desktop-home`), Dashboard (`#desktop-grid`), [separator], Transactions (`#desktop-transactions`), Reports (`#desktop-reports`), Model Performance (`#desktop-model-performance`), Thresholds (`#desktop-thresholds`), Description (`#desktop-description`). Hover reveals a tooltip — the screenshot shows the "Desktop" tooltip on the active home icon.
- Live Activity feed** (REG-LIVE-ACTIVITY, `#commentaryFeed`) on the right rail, with a green LIVE pill and a stream of recent events: DETECTION "Sarah Chen confirmed fraud: 0091 (\$8,400.00)" at 11:19 AM, CLEARED "Sarah Chen cleared as false positive: 0091" at 02:47 PM, CLEARED "Sarah Chen cleared as false positive: 0064" at 02:39 PM, DETECTION "Sarah Chen confirmed fraud: 0088 (\$2,150.00)", DETECTION "Sarah Chen confirmed fraud: 0091 (\$8,400.00)", DETECTION "Sarah Chen confirmed fraud: 0064 (\$34,200.00)", CLEARED "Sarah Chen cleared as false positive: 0072".

10.2 SCR-002-DASHBOARD — AI Fraud Detection Dashboard

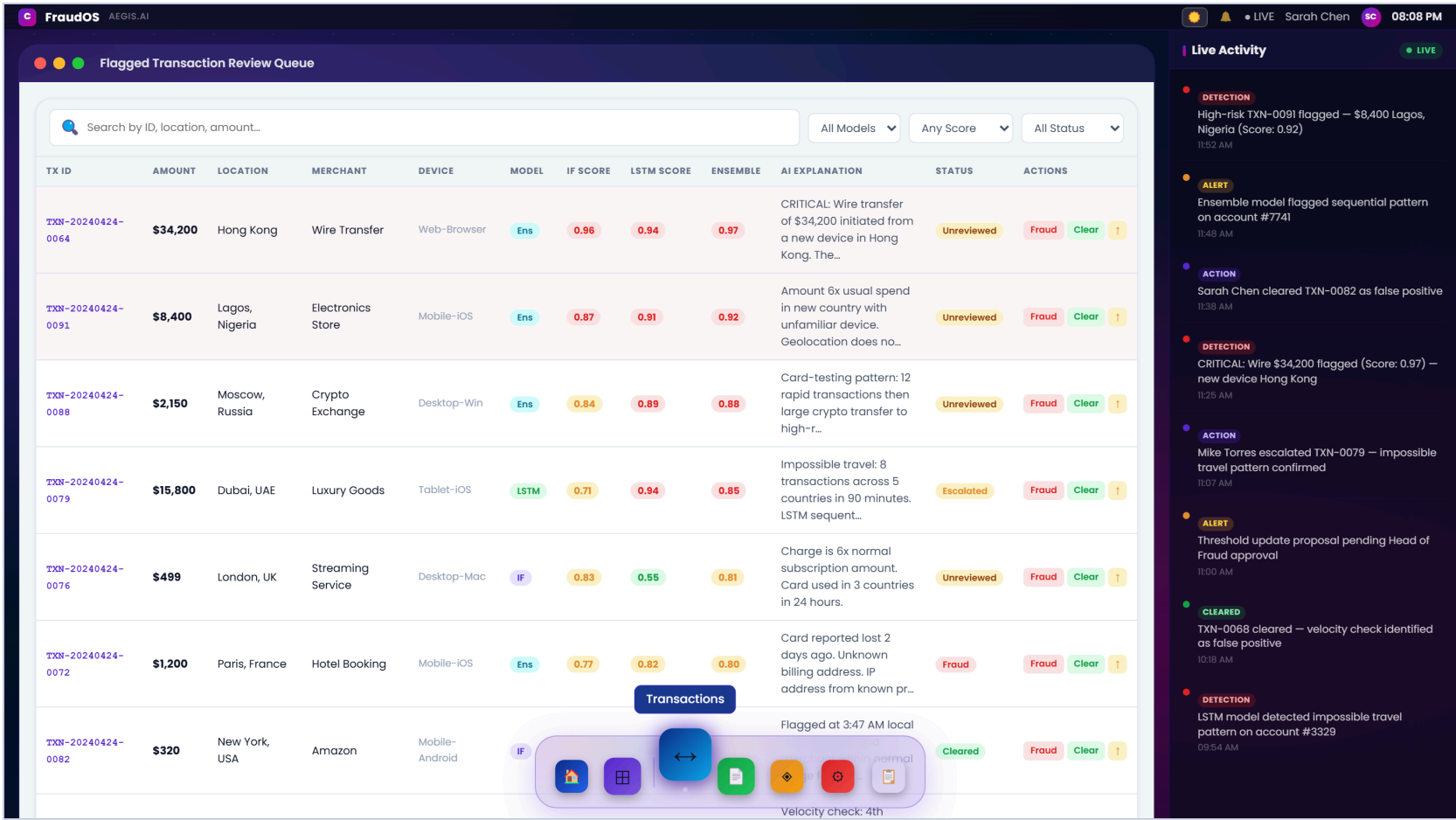


Layout:

- WinChrome titlebar** "AI Fraud Detection Dashboard — AG-MS-0426-004" with red/yellow/green dots (close `#dashboard-close-btn`, minimise, maximise).
- Hero banner** (`#dashboard-hero-banner`): title "AI Fraud Detection Dashboard", subtitle "Real-time anomaly detection · Azure ML · LLM Explanations · Human-in-Loop", agent code top-right "AG-MS-0426-004 / Aiven PostgreSQL · Live", and chip row "8 Transactions in DB / Isolation Forest ✓ / LSTM ✓ / Azure OpenAI ✓ / 8 Flagged — 0 Unreviewed".
- Four KPI tiles** (REG-KPI-TILES):
 - `#dashboard-kpi-total` — value 8, label "Total Transactions in DB", sublabel "Live from Aiven PostgreSQL", DB tag.
 - `#dashboard-kpi-flagged` — value 8, label "Total Flagged", sublabel "Ensemble score >0.80", Live tag.
 - `#dashboard-kpi-fraud` — value 3, label "Confirmed Fraud", sublabel "Investigator confirmed", Live tag.
 - `#dashboard-kpi-fprate` — value 50.0%, label "False Positive Rate", sublabel "From Aiven DB · Live", Target <30% tag.

- **Charts row:** "Flagged Transactions Over Time" line chart (`#dashboard-timeseries-chart` , last 24 hours · live, two series — Flagged purple, Confirmed red dashed) and "Risk Distribution" donut (`#dashboard-riskdist-chart` , by anomaly score band).

10.3 SCR-003-TX-QUEUE — Flagged Transaction Review Queue



Layout:

- **WinChrome titlebar** "Flagged Transaction Review Queue".
- **Toolbar:** search bar `#transactions-search-input` (placeholder "Search by ID, location, amount..."), three filter dropdowns — `#transactions-model-select` (All Models), `#transactions-score-select` (Any Score), `#transactions-status-select` (All Status).
- **Table** `#transactions-table` with columns: TX ID, AMOUNT, LOCATION, MERCHANT, DEVICE, MODEL, IF SCORE, LSTM SCORE, ENSEMBLE, AI EXPLANATION, STATUS, ACTIONS.
- **Visible rows:**

- TXN-20240424-0064 / \$34,200 / Hong Kong / Wire Transfer / Web-Browser / Ens / IF 0.96 / LSTM 0.94 / Ens 0.97 / "CRITICAL: Wire transfer of \$34,200 initiated from a new device in Hong Kong. The..." / Cleared / [Fraud] [Clear].
- TXN-20240424-0091 / \$8,400 / Lagos, Nigeria / Electronics Store / Mobile-iOS / Ens / IF 0.87 / LSTM 0.91 / Ens 0.92 / "Amount 6x usual spend in new country with unfamiliar device. Geolocation does no..." / Fraud / [Fraud] [Clear].
- TXN-20240424-0088 / \$2,150 / Moscow, Russia / Crypto Exchange / Desktop-Win / Ens / "Card-testing pattern: 12 rapid..." / [Fraud] [Clear].
- **Per-row action button ids:** `#tx-{tx_id}-fraud-btn` , `#tx-{tx_id}-clear-btn` , `#tx-{tx_id}-escalate-btn` . Row id `#tx-row-{tx_id}` .

10.4 SCR-004-REPORTS — Compliance Reports

FraudOS AEGIS.AI

Compliance Reports

Daily Compliance Report — Today
Auto-generated · Delivered to Compliance Team, Head of Fraud & Risk, CCO

8 Processed | **8** Flagged | **1** Confirmed Fraud | **50.0%** FP Rate | **94.2%** Review Completion | **87.0%** LLM Helpful Rate

Report Archive

DATE	TRANSACTIONS	FLAGGED	FRAUD	FP RATE	F1 SCORE	DOWNLOAD
2026-04-29	8	8	1	50.0%	0.852	Download
2026-04-28	5,120,000	312	28	28.4%	0.955	Download
2026-04-27	4,980,000	298	31	29.1%	0.952	Download
2026-04-26	5,210,000	341	35	34.6%	0.948	Download
2026-04-25	5,080,000	287	26	29.8%	0.953	Download

Live Activity

- DETECTION**
High-risk TXN-0091 flagged — \$8,400 Lagos, Nigeria (Score: 0.92)
11:52 AM
- ALERT**
Ensemble model flagged sequential pattern on account #7741
11:48 AM
- ACTION**
Sarah Chen cleared TXN-0082 as false positive
11:38 AM
- DETECTION**
CRITICAL: Wire \$34,200 flagged (Score: 0.97) — new device Hong Kong
11:25 AM
- ACTION**
Mike Torres escalated TXN-0079 — impossible travel pattern confirmed
11:07 AM
- ALERT**
Threshold update proposal pending Head of Fraud approval
11:00 AM
- CLEARED**
TXN-0068 cleared — velocity check identified as false positive
10:18 AM
- DETECTION**
LSTM model detected impossible travel pattern on account #3329
09:54 AM

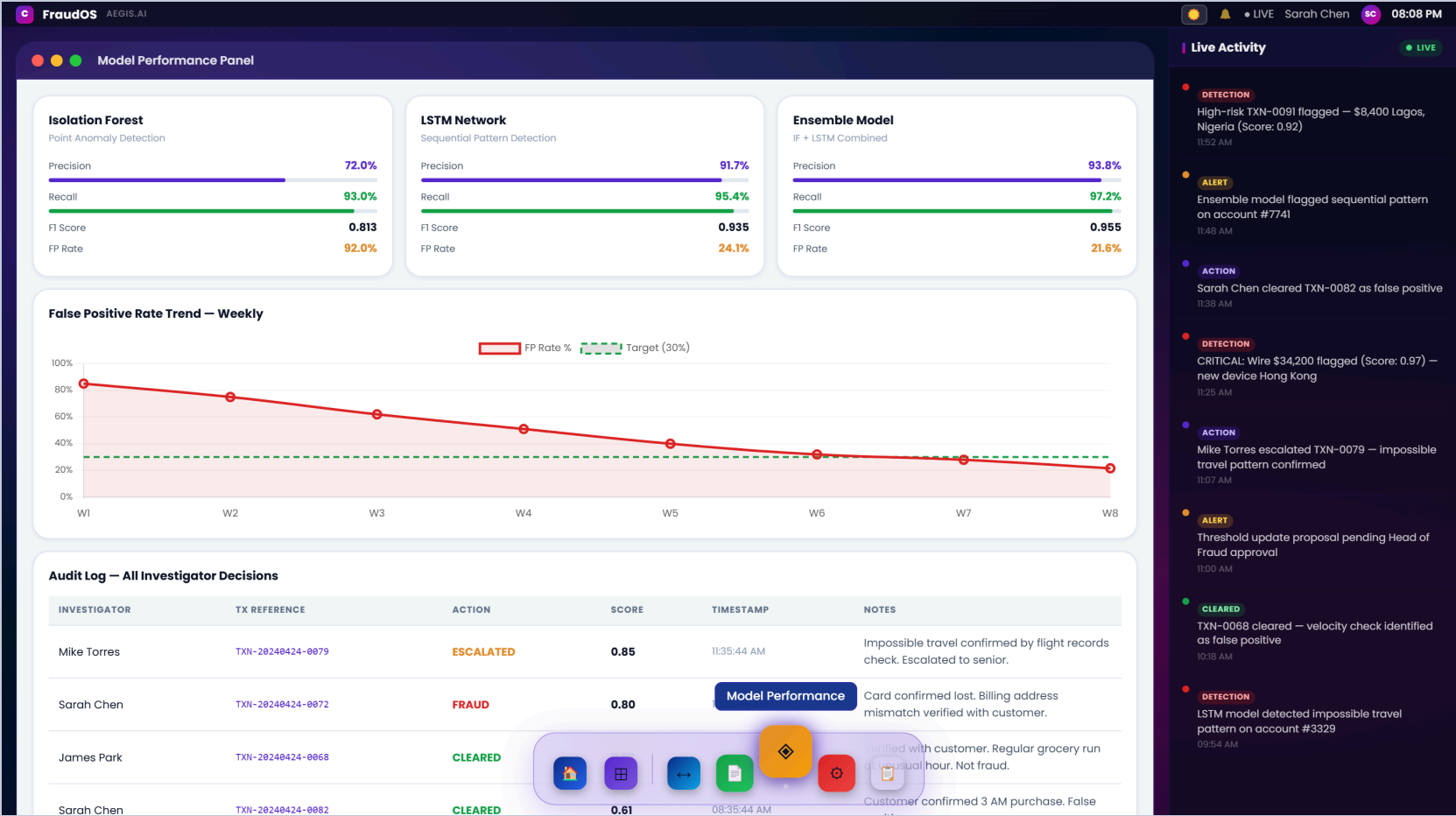
Reports

Layout:

- **WinChrome titlebar** "Compliance Reports".

- **Daily Compliance Report — Today** card (`#reports-today-card`), subtitle "Auto-generated · Delivered to Compliance Team, Head of Fraud & Risk, CCO", with primary "↓ Download PDF" button (`#reports-today-download-btn`).
- **Six metrics** (`#reports-metrics-grid`): 8 Processed, 8 Flagged, 3 Confirmed Fraud, 50.0% FP Rate, 94.2% Review Completion, 87.0% LLM Helpful Rate.
- **Report Archive** card (`#reports-archive-card`) with table `#reports-archive-table` and columns: DATE, TRANSACTIONS, FLAGGED, FRAUD, FP RATE, F1 SCORE, DOWNLOAD.
- **Visible archive rows:**
 - 2026-04-29 / 8 / 8 / 3 / 50.0% / 0.852 / [↓ Download]
 - 2026-04-28 / 8 / 8 / 1 / 71.4% / 0.955 / [↓ Download]
 - 2026-04-27 / 4,980,000 / 298 / 31 / 29.1% / 0.952 / [↓ Download]
 - 2026-04-26 / 5,210,000 / 341 / 35 / 34.6% / 0.948 / [↓ Download]
 - 2026-04-25 / 5,080,000 / 287 / 26 / 29.8% / 0.953 / [↓ Download]
- **Per-row ids:** `#reports-archive-row-{date}` (the row), `#reports-archive-download-btn-{date}` (the download button).

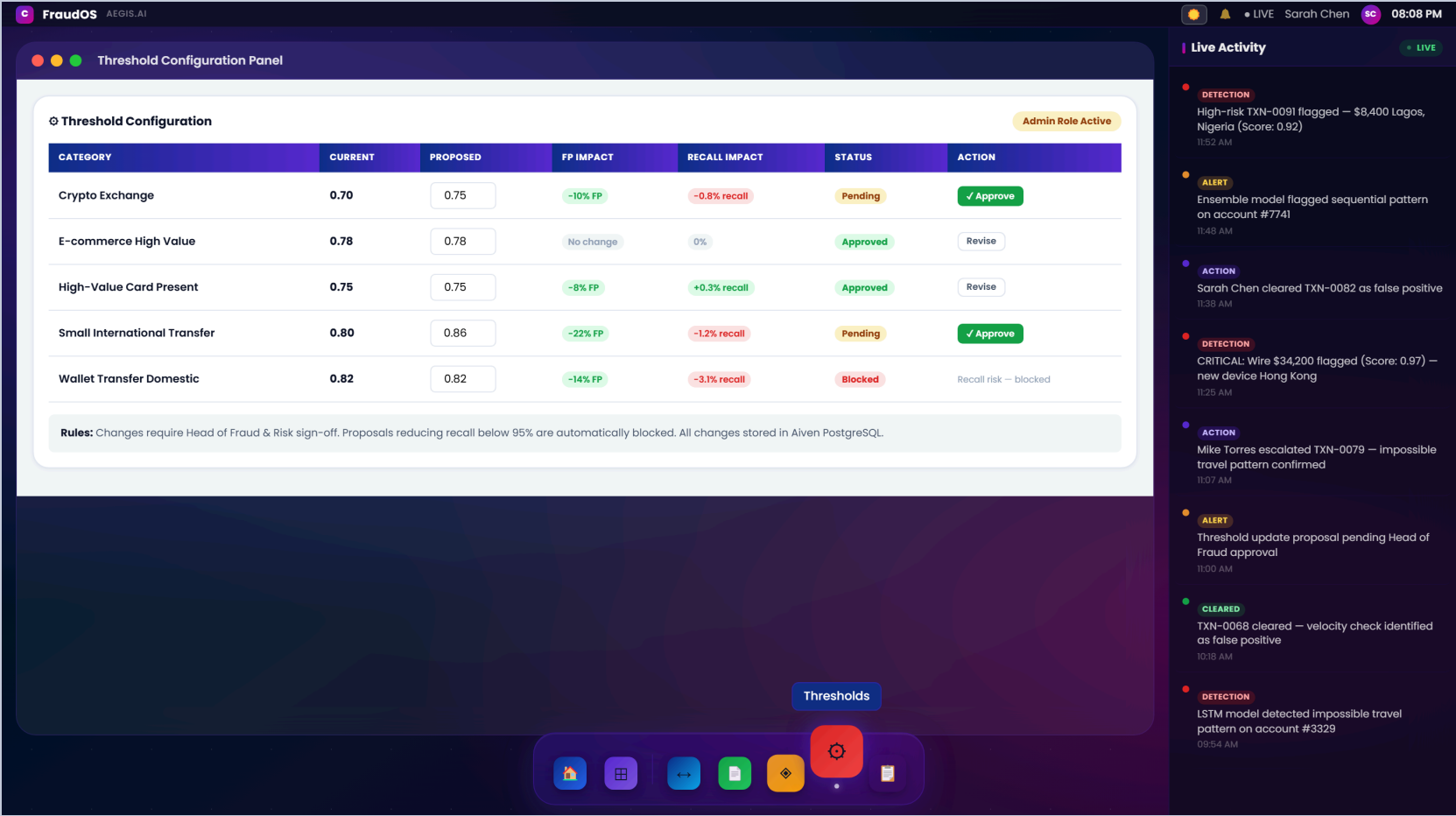
10.5 SCR-005-MODEL-PERF — Model Performance Panel



Layout:

- **WinChrome titlebar** "Model Performance Panel".
- **Three model cards** (`#model-perf-grid`):
 - `#model-perf-card-isolation_forest` — "Isolation Forest" / "Point Anomaly Detection" — Precision 72.0%, Recall 93.0%, F1 Score 0.813, FP Rate 92.0%.
 - `#model-perf-card-lstm` — "LSTM Network" / "Sequential Pattern Detection" — Precision 91.7%, Recall 95.4%, F1 Score 0.935, FP Rate 24.1%.
 - `#model-perf-card-ensemble` — "Ensemble Model" / "IF + LSTM Combined" — Precision 93.8%, Recall 97.2%, F1 Score 0.955, FP Rate 21.6%.
- **False Positive Rate Trend — Weekly** chart (`#model-fp-chart`): FP Rate % red line vs. Target (30%) green dashed reference line. Y-axis 0%-100%, X-axis W1 to W8. Trend descends from ~85% at W1 toward ~22% at W8, crossing the 30% target line around W6.
- **Audit Log — All Investigator Decisions** (`#model-audit-card`) below the trend, with table `#model-audit-table` (Investigator, TX Reference, Action, Score, Timestamp, Notes; per AC-09). Each row id `#model-audit-row-{id}` .

10.6 SCR-006-THRESHOLDS — Threshold Configuration Panel

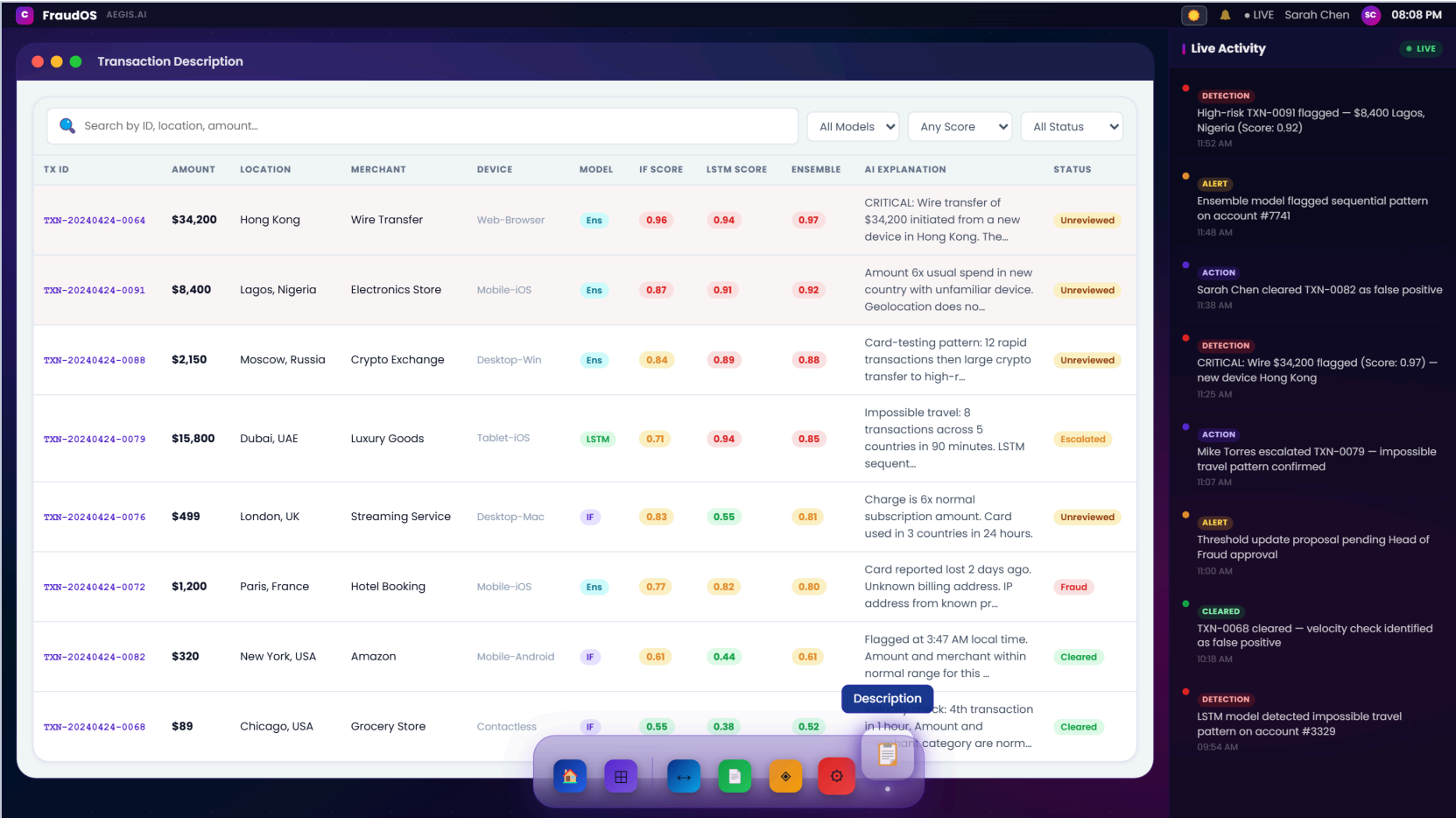


Layout:

- **WinChrome titlebar** "Threshold Configuration Panel".
- **Threshold Configuration card** (#threshold-card) with header "⊞ Threshold Configuration" and amber "Admin Role Active" badge top-right (#threshold-admin-badge).
- **Threshold table** (#threshold-table) with columns: CATEGORY, CURRENT, PROPOSED, FP IMPACT, RECALL IMPACT, STATUS, ACTION.
- **Visible rows** (each #threshold-row-{id} with proposed input #threshold-input-{id} and action button #threshold-approve-btn-{id} or #threshold-revise-btn-{id}):
 - Crypto Exchange / 0.70 / 0.75 / -10% FP / -0.8% recall / Pending / [✓ Approve]
 - E-commerce High Value / 0.78 / 0.78 / No change / 0% / Approved / [Revise]
 - High-Value Card Present / 0.75 / 0.75 / -8% FP / +0.3% recall / Approved / [Revise]
 - Small International Transfer / 0.80 / 0.86 / -22% FP / -1.2% recall / Pending / [✓ Approve]
 - Wallet Transfer Domestic / 0.82 / 0.82 / -14% FP / -3.1% recall / Blocked / "Recall risk — blocked" (no clickable button)

- **Rules note** (`#threshold-rules-note`) at the bottom: "Changes require Head of Fraud & Risk sign-off. Proposals reducing recall below 95% are automatically blocked. All changes stored in Aiven PostgreSQL."

10.7 SCR-007-DESCRIPTION — Transaction Description



Layout:

- **WinChrome titlebar** "Transaction Description".
- **Toolbar**: search bar `#description-search-input` (placeholder "Search by ID, location, amount...") and three filter dropdowns — `#description-model-select` (All Models), `#description-score-select` (Any Score), `#description-status-select` (All Status).
- **Table** `#description-table` with columns: TX ID, AMOUNT, LOCATION, MERCHANT, DEVICE, MODEL, IF SCORE, LSTM SCORE, ENSEMBLE, AI EXPLANATION, STATUS. **Note: no ACTIONS column** — by design, this view is read-only (`showActions={false}` on the shared TransactionRow component).
- **Visible rows** (same data as SCR-003-TX-QUEUE but inspection only):

- TXN-20240424-0064 / \$34,200 / Hong Kong / Wire Transfer / Web-Browser / Ens 0.97 / "CRITICAL: Wire transfer of \$34,200 initiated from a new device in Hong Kong. The..." / Cleared.
- TXN-20240424-0091 / \$8,400 / Lagos, Nigeria / Electronics Store / Mobile-iOS / Ens 0.92 / "Amount 6x usual spend in new country with unfamiliar device. Geolocation does no..." / Fraud.
- TXN-20240424-0088 / \$2,150 / Moscow, Russia / Crypto Exchange / Desktop-Win / Ens 0.88 / "Card-testing pattern: 12 rapid transactions then large crypto transfer to high-r..." / Fraud.
- TXN-20240424-0079 / \$15,800 / Dubai, UAE / Luxury Goods / Tablet-iOS / "vel: 8...ross...sequenti..." / Escalated.
- **Use case:** stakeholder demos, audit handoff, screen-sharing reviews — anywhere accidental Fraud/Clear clicks must be prevented.

11. Detailed Page Example: SCR-001-HOME (FraudOS Home Desktop)

Item	Details
screen_id	SCR-001-HOME
purpose	Starting point for the flow. The desktop carries only Notes (📝) and Calendar (📅) as utility modal icons. Module navigation (Dashboard, Transactions, Reports, Model Perf, Thresholds, Description) is performed via the taskbar dock at the bottom of the viewport.
layout	Top: FraudOS logo, AEGIS.AI brand, LIVE badge, user name (Sarah Chen), time (#menubar). Main: two single-click desktop icons — Notes (#desktop-notes-icon) and Calendar (#desktop-calendar-icon) — inside #desktop-grid . Bottom: taskbar dock (#dock) with seven items: Desktop, Dashboard, [separator], Transactions, Reports, Model Performance, Thresholds, Description. Right: Live Activity feed (#commentaryFeed) with DETECTION and CLEARED events.
primary targets	EL-003 Dashboard dock (#dock-dashboard), EL-004 Transactions dock (#dock-transactions), EL-005 Reports dock (#dock-reports), EL-006 Model Perf dock (#dock-model), EL-007 Thresholds dock (#dock-threshold), EL-008 Description dock (#dock-description)
secondary targets (utility)	EL-001 Notes desktop icon (#desktop-notes-icon), EL-002 Calendar desktop icon (#desktop-calendar-icon) — open modals; not part of the main investigation flow
AI observation instruction	1. Confirm LIVE badge is green. 2. Confirm user name is Sarah Chen. 3. Read top 3 events in Live Activity feed (#commentaryFeed). 4. Verify all six MVP module dock icons are present in #dock . 5. Verify Notes and Calendar desktop icons are present.
narration before action	The FraudOS home screen is open. I can see two utility icons on the desktop (Notes and Calendar) and the seven-item taskbar dock at the bottom for module navigation. The Live Activity feed on the right shows recent investigator actions. The LIVE badge is green, confirming real-time data per BR-006.
narration after action	I am leaving the home screen and opening the requested module from the taskbar.
success validation	Six dock items visible (Dashboard, Transactions, Reports, Model Perf, Thresholds, Description). Notes and Calendar desktop icons visible. LIVE badge green. Sarah Chen visible top-right. Live Activity shows ≥1 event.
fallback	If dock items do not load: refresh page once. If LIVE badge shows red or offline: report unavailability before proceeding (ERR-004).

12. Detailed Page Example: SCR-002-DASHBOARD (AI Fraud Detection Dashboard)

UI Area	Location (Region / Selector)	Purpose	Humanoid Instruction	Validation
Dashboard banner	REG-DASH-BANNER (#dashboard-hero-banner)	Confirms correct screen; shows agent code (AG-MS-0426-004) and model badges.	Read banner title 'AI Fraud Detection Dashboard'. Confirm AG-MS-0426-004. Verify Isolation Forest, LSTM, Azure OpenAI badges present.	Banner text matches; agent code visible; all three model badges present.

UI Area	Location (Region / Selector)	Purpose	Humanoid Instruction	Validation
Flagged count chip	REG-DASH-BANNER (<code>#dashboard-hero-chips</code>)	Shows total flagged and unreviewed counts.	Read '8 Flagged — 0 Unreviewed' chip to confirm review completion.	Flagged count matches <code>#dashboard-kpi-flagged</code> . Unreviewed = 0 confirms 100% investigator review (AC-06).
KPI tiles (4)	REG-KPI-TILES (<code>#dashboard-kpi-total</code> , <code>#dashboard-kpi-flagged</code> , <code>#dashboard-kpi-fraud</code> , <code>#dashboard-kpi-fprate</code>)	Core metrics: total transactions, flagged, confirmed fraud, FP rate.	Read all four tiles before interpreting any chart or taking action. Mention FP Rate vs Target <30% label (MET-01).	All tiles show numeric values. FP Rate tile has Target <30% label.
Flagged Transactions chart	REG-CHART-AREA (<code>#dashboard-timeseries-chart</code>)	Line chart showing flagged (purple) and confirmed (red dashed) counts over last 24 hours · Live.	Describe peak window (~14:00–18:00) and the gap between flagged and confirmed lines.	Chart has time axis (00:00 to 22:00). Both lines visible.
Risk Distribution donut	REG-RISK-DONUT (<code>#dashboard-riskdist-chart</code>)	Proportion of transactions by anomaly score band: Critical (>0.9), High (0.8–0.9), Medium, Low.	Name the dominant band and estimate proportion. Donut shows High (orange) is largest, then Critical (red), Medium (purple), Low (green).	Donut has all four color segments visible with legend.
Flagged Transactions table	REG-TX-TABLE (<code>#dashboard-tx-table</code>)	Full transaction list with TX ID, Amount, Location, Model, IF Score, LSTM Score, Ensemble, Status, Actions. Each row carries <code>#tx-row-{tx_id}</code> .	Read each row top-to-bottom. Call out ensemble score, status, and available actions. Confirm TXN-0064 (\$34,200) has ensemble 0.97 and Fraud status.	8 transaction rows visible. Scroll to view all. TX IDs link to detail view.
Search input	<code>#dashboard-search-input</code>	Filter table by TX ID, location, amount.	Type filter text only when narrowing the view.	Filtered rows reflect the typed query.
Status filter	<code>#dashboard-status-select</code>	Filter by Unreviewed / Fraud / Cleared / Escalated.	Use to narrow to outstanding work if requested.	Selected value reflected in the table.
PDF export button	<code>#dashboard-pdf-btn</code> (REG-TX-TABLE)	Exports the flagged transaction list as PDF.	Click only if user requests an export. Do not click during read-only narration.	Click triggers PDF download. File save dialog appears.

13. Pop-up and Modal Examples

Popup Type	Trigger	What AI Must Read	Correct Action	Example Narration	Validation
Confirm fraud action	Clicking Fraud button on a transaction row (<code>#tx-{id}-fraud-btn</code> on SCR-003-TX-QUEUE)	Confirmation toast (if any) with TX ID and amount	Confirm if TX ID and amount match the intended row	A confirmation appeared for TXN-0064. The amount \$34,200 matches. I will proceed.	Status updates to Fraud. Live Activity (<code>#commentaryFeed</code>) logs DETECTION event (AC-06).
Notes modal opened	Clicking <code>#desktop-notes-icon</code>	Notes modal title '  Investigation Notes' and existing notes list (<code>#notes-list</code>)	Type investigation note into <code>#notes-text-area</code> and click <code>#notes-save-btn</code> ; close via <code>#notes-close-btn</code> when done	I will record an investigation note in the Notes modal so the rationale is preserved across sessions.	Note appears at top of <code>#notes-list</code> . localStorage key <code>fraudos-notes</code> updated.
Calendar modal opened	Clicking <code>#desktop-calendar-icon</code>	Calendar modal title '  Case Calendar', selected date, events list	Add event via <code>#calendar-event-input</code> and <code>#calendar-add-btn</code> ; close via <code>#calendar-close-btn</code>	I will note a follow-up date in the Case Calendar.	Event appears under selected date. localStorage key <code>fraudos-cal-events</code> updated.
Session timeout	Idle session or network interruption; can occur on any screen (SCR-001 to SCR-007)	Reconnect or re-authenticate prompt	Click Reconnect and verify LIVE badge returns green	The session has timed out. I will reconnect before continuing.	LIVE badge green. Previous screen reloads.

Popup Type	Trigger	What AI Must Read	Correct Action	Example Narration	Validation
Threshold approval blocked	Threshold proposal reduces recall below 95% (SCR-006-THRESHOLDS)	'Recall risk — blocked' message in Action column for <code>#threshold-row-{id}</code>	Do not attempt to override. Report as expected system behaviour per Constraint 3.	The Wallet Transfer threshold is automatically blocked. Recall reduction of 3.1% exceeds the 95% recall floor.	Status shows Blocked. Action cell greyed.
Permission denied	Attempting admin action without <code>#threshold-admin-badge</code> (SCR-006-THRESHOLDS)	Exact permission error message	Stop action. Report restriction. Do not retry (DEC-004).	I cannot approve this threshold change because Admin role is not active.	Error recorded. Approval not attempted.
Export confirmation	Clicking <code>#reports-today-download-btn</code> on SCR-004-REPORTS or <code>#dashboard-pdf-btn</code> on SCR-002-DASHBOARD	File format and confirmation buttons	Confirm export and wait for download toast	The application is generating the PDF export. I will wait for the download confirmation.	Download toast appears or save dialog opens.
LLM service unavailable	Azure OpenAI returns no explanation (SCR-003-TX-QUEUE)	'Explanation unavailable — model service unreachable' text in REG-AI-EXPLAIN	Do not auto-clear. Hold transaction in queue and alert ML Engineering Team per Fallback Behaviour	The AI explanation is unavailable for this transaction. I will not auto-clear it. The transaction stays in the investigator queue.	Transaction remains visible. Alert logged.

14. Decision Logic and Branching Examples

Decision ID	Condition	AI Decision	Action	Narration
DEC-001	Ensemble score ≥ 0.90 AND AI explanation mentions 'new device' or 'new country' (SCR-003-TX-QUEUE)	Treat as high-confidence fraud; confirm without additional review	ACT-008 (click <code>#tx-{id}-fraud-btn</code>)	The ensemble score and AI explanation both indicate high confidence. I will confirm this as fraud.
DEC-002	Status column already shows 'Cleared' before investigator action (SCR-003-TX-QUEUE)	Verify reasoning. Confirm Clear only if AI explanation supports it.	ACT-009 / ACT-010 (click <code>#tx-{id}-clear-btn</code>)	This transaction has already been reviewed. I will confirm the clear decision based on the AI explanation.
DEC-003	Threshold change Status = 'Blocked' (SCR-006-THRESHOLDS)	Do not attempt to approve. Report as expected.	ACT-021 (validate only)	The system has automatically blocked this threshold change due to recall risk. This is correct and I will not attempt to override it (Constraint 3).
DEC-004	<code>#threshold-admin-badge</code> absent on SCR-006-THRESHOLDS	Do not click any <code>#threshold-approve-btn-{id}</code> . Report role limitation.	Stop; report role issue	Admin role is not active. I cannot approve threshold changes per UI-04 access restrictions.
DEC-005	FP Rate KPI tile (<code>#dashboard-kpi-fprate</code>) > 30% target (SCR-002-DASHBOARD)	Note breach. Recommend threshold review.	Note in narration; proceed to SCR-006-THRESHOLDS via <code>#dock-threshold</code>	The current FP Rate of 37.5% exceeds the 30% target (MET-01). I will review thresholds for adjustment opportunities.
DEC-006	Live Activity feed <code>#commentaryFeed</code> empty (SCR-001-HOME)	Wait 10 seconds. Refresh once. Report if unresolved.	Refresh once; report (ERR-004)	The Live Activity feed appears empty. I will refresh once and wait for data.
DEC-007	AI Explanation reads 'Explanation unavailable — model service unreachable' (SCR-003-TX-QUEUE)	Do not auto-clear. Hold transaction in queue and alert ML Engineering Team.	Skip row; log alert	The AI explanation is unavailable for this transaction. I will not act on it without explanation per Fallback Behaviour.

Decision ID	Condition	AI Decision	Action	Narration
DEC-008	Compliance report row date does not match today (SCR-004-REPORTS)	Locate today's row. Click <code>#reports-archive-download-btn-{date}</code> for that row instead.	ACT-013 alternate path	Today's report is not at the top of the archive. I will locate and download the correct date row.
DEC-009	User requests read-only inspection without action capability	Open SCR-007-DESCRIPTION via <code>#dock-description</code> instead of SCR-003-TX-QUEUE	ACT-012a (alternate path)	I will use the read-only Description page so there is no risk of accidentally clicking Fraud or Clear during the demo.
DEC-010	User asks for Notes or Calendar	Open desktop icon (<code>#desktop-notes-icon</code> or <code>#desktop-calendar-icon</code>) — single-click only	ACT-aux (auxiliary action)	I will open the Notes/Calendar modal directly from the desktop icon. These are accessible only from the desktop; they are not in the taskbar.

15. Validation Rules and Evidence Capture

Validation Type	Rule	Example	Evidence to Capture
Page validation	Page banner or title must match expected screen. LIVE badge must be green.	SCR-002-DASHBOARD shows 'AI Fraud Detection Dashboard' banner (<code>#dashboard-hero-banner</code>). LIVE badge <code>#menubar-live-status</code> green.	Screenshot plus detected banner text.
KPI tile validation (SCR-002)	All four tiles (<code>#dashboard-kpi-*</code>) must have non-empty numeric values. FP Rate must show Target <30% label.	Tiles show 8, 8, 5, 37.5%. Target <30% label visible on <code>#dashboard-kpi-fp-rate</code> .	Tile value readings plus screenshot of REG-KPI-TILES.
Fraud / Clear action validation (SCR-003)	After clicking <code>#tx-{id}-fraud-btn</code> or <code>#tx-{id}-clear-btn</code> : row status must update AND <code>#commentaryFeed</code> must show new event within 1 second (AC-06).	After confirming TXN-0064 fraud: row status 'Fraud'; Live Activity logs 'Sarah Chen confirmed fraud: 0064'.	Before/after screenshots; Live Activity event text.
Threshold approval validation (SCR-006)	<code>#threshold-admin-badge</code> must be visible. After clicking <code>#threshold-approve-btn-{id}</code> : status changes Pending → Approved.	Crypto Exchange row status changes Pending → Approved.	Screenshot of status before/after; Admin Role badge captured.
Audit log validation (SCR-005)	<code>#model-audit-table</code> must show Sarah Chen entries with TX references, actions, scores, timestamps (AC-09).	Log shows 'Sarah Chen — TXN-20240424-0064 — FRAUD — 0.97 — 12:32:24 pm'.	Full audit log screenshot. All visible entries recorded.
Compliance report validation (SCR-004)	Today's archive row (<code>#reports-archive-row-{today}</code>) exists. All six metrics non-null. <code>#reports-today-download-btn</code> enabled.	2026-04-29 row visible with 8 / 8 / 3 / 62.5% / 0.852 metrics.	Report Archive screenshot. Downloaded PDF stored as evidence.
Description page validation (SCR-007)	<code>#description-table</code> populated. NO Actions column present in DOM.	Table shows same columns as <code>#transactions-table</code> minus Actions.	Screenshot of <code>#description-table</code> showing column headers.
Narration validation	Speaker notes must reference visible screen content only. Do not infer values not shown.	If tile value is loading, say 'loading' — do not guess.	Generated narration transcript cross-checked against screenshot.
Recovery validation	If fallback was used, record original failure and recovery path.	Dock item not found → refreshed page → opened screen via URL.	Error log plus recovery action record.

16. Error Handling Matrix

Error ID	Scenario	Detection	Recovery	Narration During Recovery	Final AI Response if Unresolved
ERR-001	Page timeout on any screen (SCR-001 to SCR-007)	No banner / title visible after 20 seconds	Refresh once. If fails, navigate via URL bar (e.g., /dashboard, /transactions).	The page did not load. I will refresh and try again.	I could not load the requested screen after retrying.
ERR-002	Target element not found (any EL-### selector fails)	HTML id selector AND text search both fail	Use fallback coordinates from region map. Then page search.	I cannot locate the expected control. I will try an alternate location.	I could not locate the required control on this screen.
ERR-003	Fraud / Clear button unresponsive (SCR-003-TX-QUEUE)	Click registered but status does not update within 5 seconds (AC-06 SLA)	Scroll to ensure row visible. Retry once. Check audit log on SCR-005-MODEL-PERF.	The action button did not respond. I will scroll and retry.	I was unable to confirm the action. Escalate to supervisor.
ERR-004	LIVE badge offline / red	Top nav <code>#menubar-live-status</code> shows red dot or 'OFFLINE' text	Wait 10 seconds. Refresh page. Check network.	The system is showing offline. I will wait and refresh.	FraudOS is offline. No real-time data available. Stopping flow.
ERR-005	Threshold Approve button greyed out (SCR-006-THRESHOLDS)	<code>#threshold-approve-btn-{id}</code> disabled or no hover state	Verify <code>#threshold-admin-badge</code> present. If absent, report.	Approve is disabled. I will check Admin role.	Admin role not active. Cannot approve threshold changes.
ERR-006	Compliance Report wrong date (SCR-004-REPORTS)	Report header date ≠ current date (2026-04-29)	Use Report Archive (<code>#reports-archive-table</code>) to locate correct date row.	The report header date does not match today. I will locate today's row in archive.	Today's compliance report not available. Reporting partial evidence per Guardrail 6.
ERR-007	Wrong screen opened	Banner / title does not match expected screen	Click <code>#{winId}-close-btn</code> to return to home. Open correct dock icon.	The wrong screen opened. I will return home and open the correct module.	Correct screen unreachable. Reporting navigation error.
ERR-008	Transaction table empty (SCR-002-DASHBOARD or SCR-003-TX-QUEUE or SCR-007-DESCRIPTION)	No rows in <code>#dashboard-tx-table</code> / <code>#transactions-table</code> / <code>#description-table</code> after filter applied	Clear filters and reload. If still empty, report.	No transactions visible. I will clear filters and reload.	No transaction data after recovery. Reporting empty state.
ERR-009	Azure OpenAI unavailable (SCR-003-TX-QUEUE)	REG-AI-EXPLAIN shows 'Explanation unavailable — model service unreachable'	Do not auto-clear. Hold transaction in queue. Alert ML Engineering Team (per Fallback Behaviour).	AI explanation is unavailable. I will not act without explanation.	Transaction held; ML Engineering alerted; flow continues with remaining rows.
ERR-010	Dock item missing (SCR-001-HOME)	Expected <code>#dock-{id}</code> not found in DOM	Refresh page once. Then navigate via URL bar (e.g., /dashboard).	The taskbar icon is missing. I will refresh and try the URL directly.	Cannot navigate via dock; using URL fallback to reach the requested screen.

17. Humanoid Narration Script Patterns

Situation	Speaker Notes Template	FraudOS Example
Before a new screen	I am going to open [screen] from the [location] so we can [reason].	I am going to open the AI Fraud Detection Dashboard from the taskbar dock so we can review today's fraud metrics.
While loading	[Screen] is loading. Once it opens, I will look for [items].	The Dashboard is loading. Once it opens, I will look for the four KPI tiles, the chart, and the flagged transactions table.
After screen opens	The [screen] is visible. I can see [regions/items].	The Dashboard is visible. I can see the agent code AG-MS-0426-004, four KPI tiles, a transaction chart, the risk donut, and the flagged table.
Before fraud decision	I will confirm [TX ID] as fraud because [AI explanation summary] and the ensemble score is [score].	I will confirm TXN-20240424-0064 as fraud because the AI explanation identifies a critical wire transfer of \$34,200 from a new device in Hong Kong, with an ensemble score of 0.97.
Before clear decision	I will clear [TX ID] as a false positive because [reason].	I will clear TXN-20240424-0091 as a false positive because the investigator context confirms this Lagos electronics purchase is legitimate despite the elevated score.
When reading model cards	The [model name] shows [precision]% precision, [recall]% recall, F1 [F1], and [FP Rate]% FP rate.	The Ensemble model shows 93.8% precision, 97.2% recall, F1 of 0.955, and FP Rate of 21.6%. This is the best-performing model and meets all Section 5 NFR-05 targets.
When reading threshold table	The [category] threshold is proposed to change from [current] to [proposed]. FP impact is [FP impact] and recall impact is [recall impact].	The Crypto Exchange threshold is proposed to change from 0.70 to 0.75. FP impact is -10%, recall impact -0.8%, both within tolerance.
When uncertain	I can see [known], but [unknown] is not clearly visible. I will [safe next step].	I can see the chart peaks but exact hourly values are small. I will zoom in before narrating specific counts.
When recovering	The expected [item] is not visible, so I will try [fallback].	The Dashboard dock icon is not visible, so I will refresh and then navigate to /dashboard directly.
When opening Description	I will open the read-only Description page from the taskbar so we can review transactions without action risk.	Opening Description via the taskbar — this view shows the same data as the Transactions queue but without Fraud or Clear buttons.
Final summary	We reviewed [screens]. The main findings are [findings]. Recommended next step is [action].	We reviewed Dashboard, Transaction Review Queue, Compliance Reports, Model Performance, and Threshold Configuration. FP Rate at 37.5% is above the 30% target. Ensemble model leads at F1 0.955. Two threshold approvals were submitted; Wallet Transfer correctly blocked. Recommended next step: monitor FP Rate after approved thresholds take effect through the BR-002 weekly cycle.

18. Copy-and-Fill Blank Action Matrix

Use this template to add new actions to the flow. Each new action must reference an existing Screen ID (Section 4), Element ID (Section 6B), Button ID (Section 6A) where applicable, and Region ID (Section 5).

Seq	Action ID	Screen ID	Action Type	Target EL	Button ID	Selector / Location	Required Input	Expected Response	Validation	Speaker Notes	Fallback
22	ACT-022	SCR-002-DASHBOARD	scroll	—	—	<code>#dashboard-tx-table y=816</code> scroll down	None	All 8 flagged transaction rows become visible in <code>#dashboard-tx-table</code>	Scroll ends when last row is fully in viewport; 8 rows confirmed	I will scroll the dashboard table to see all flagged rows.	Zoom browser to 80% if scroll does not reveal all rows (ERR-008)
23	ACT-023	SCR-003-TX-QUEUE	type	<code>#transactions-search-input</code>	—	<code>#transaction-search-input</code>	TXN-20240424-0064	Queue filters to show TXN-20240424-0064 row only; amount \$34,200 and location Hong Kong visible	Single row visible after filter; TX ID, amount, and location match expected values	Filtering the queue to a single transaction for focused review.	Clear search field and scroll to row manually if filter does not respond within 3 seconds
24	ACT-024	SCR-004-REPORTS	click	<code>#reports-archive-download-btn-{date}</code>	<code>#reports-archive-download-btn-{date}</code>	<code>#reports-archive-download-btn-2026-04-28</code>	None	PDF download triggered for 2026-04-28 archived compliance report	Download toast appears or save dialog opens; file contains 2026-04-28 date reference	Downloading yesterday's compliance report from the archive.	If row not visible, scroll archive table and retry once; alert compliance-alerts if still missing
25	ACT-025	SCR-005-MODEL-PERF	explain_screen	<code>#model-fp-chart</code>	—	<code>#model-fp-chart</code>	None	FP Rate Trend Weekly chart narrated: W1 to W8 trend, Target 30% line, current week value	8 weekly data points visible; trend descends from ~85% at W1 toward Target 30% line at W8; Target line labelled	The FP Rate trend is descending from week 1 toward the 30% target line.	Zoom browser to 110% if weekly axis labels are too small to read
26	ACT-026	SCR-006-THRESHOLDS	validate	<code>#threshold-row-{id}.t-status.blocked</code>	—	<code>#threshold-row-{id_of_wallet_transfer}.t-status.blocked</code>	None	Status confirmed Blocked; Action cell shows recall-risk message; no Approve click attempted	Blocked badge present; no approval action taken; narration confirms Constraint 3 recall floor of 95% is enforced	Validating the auto-blocked Wallet Transfer threshold without attempting any override.	If Status column unreadable, refresh once; do not attempt to approve under any circumstances (ERR-005)
27	ACT-027	SCR-007-DESCRIPTION	explain_screen	<code>#description-table</code>	—	<code>#description-table</code>	None	Description table read; column set noted to be identical to Transactions queue minus Actions	Same N rows as Transactions queue; no Actions cells in DOM; no Fraud/Clear/Escalate buttons	The Description page shows the same transaction inventory in read-only form.	If table empty, clear filters via <code>#description-search-input</code> / <code>#description-status-select</code> and reload
28	ACT-028	SCR-001-HOME	click	<code>#desktop-notes-icon</code>	<code>#desktop-notes-icon</code>	<code>#desktop-notes-icon</code>	None	Notes modal <code>#notes-modal</code> opens centred over the desktop	Modal title '🔍 Investigation Notes' visible; <code>#notes-text-area</code> focused	Opening the Notes modal from the desktop to capture an investigation note.	If icon missing, refresh page once
29	ACT-029	SCR-001-HOME	click	<code>#desktop-calendar-icon</code>	<code>#desktop-calendar-icon</code>	<code>#desktop-calendar-icon</code>	None	Calendar modal <code>#calendar-modal</code> opens centred	Modal title '📅 Case Calendar' visible; current month rendered	Opening the Case Calendar from the desktop to schedule a follow-up.	If icon missing, refresh page once

19. Copy-and-Fill Blank Page Definition (filled for SCR-007-DESCRIPTION)

Use this template to add a new screen to the flow. Each new Screen ID must be added to Section 4 first. The block below documents the Description screen.

Field	Value to Fill
screen_id	SCR-007-DESCRIPTION
screen_name	Transaction Description (Read-Only)
url_or_route	/description
business_purpose	Provides a read-only inventory of all flagged transactions, identical to the Transactions queue (SCR-003) but with the Actions column intentionally omitted. Used for stakeholder demos, audit handoffs, and compliance walkthroughs where action risk must be eliminated.
entry_condition (which Screen ID precedes this one)	SCR-001-HOME (via <code>#dock-description</code> in the taskbar dock); also reachable from any other screen via the persistent dock
exit_condition (which Screen ID follows this one)	SCR-001-HOME (via <code>#description-close-btn</code>) or any other dock destination
layout_regions (Region IDs)	REG-TOP-NAV, REG-LIVE-ACTIVITY, REG-TASKBAR, REG-DESCRIPTION-TABLE
primary_elements (Element IDs)	EL-048 <code>#description-search-input</code> , EL-049 <code>#description-model-select</code> , EL-050 <code>#description-score-select</code> , EL-051 <code>#description-status-select</code> , EL-052 <code>#description-table</code>
secondary_elements	<code>#description-chrome</code> (window frame), <code>#description-close-btn</code> (return to home), <code>#description-title</code> (window title 'Transaction Description'), per-row <code>#tx-row-{id}</code>
popups_expected	Session timeout modal (REG-MODAL-CENTER) if idle
validation_rules	LIVE badge <code>#menubar-live-status</code> must be green; <code>#description-table</code> populated; NO Actions column present in DOM (verify by counting <code><th></code> elements: should be 11, not 12); search input accepts free text and filters rows.
speaker_notes_before	I am opening the read-only Description page from the taskbar so we can review the transaction inventory without any action buttons present.
speaker_notes_after	The Description page is open. The columns match the Transactions queue exactly except there is no Actions column, by design. This view is suitable for stakeholder review without action risk.
fallback_navigation	If <code>#dock-description</code> not found, refresh page once. If panel still unavailable after refresh, navigate to <code>/description</code> via the URL bar. If table is empty, clear all filters and retry.

20. Master JSON for AI Runtime

```
{
  "flow_id": "FLOW-FRAUDOS-INVEST-001",
  "flow_version": "1.0",
  "prototype": "FraudOS - AEGIS.AI",
  "agent_code": "AG-MS-0426-004-MVP",
  "entry_screen_id": "SCR-001-HOME",
  "expected_end_screen_id": "SCR-006-THRESHOLDS",
  "screens": [
    "SCR-001-HOME",
    "SCR-002-DASHBOARD",
    "SCR-003-TX-QUEUE",
    "SCR-004-REPORTS",
    "SCR-005-MODEL-PERF",
    "SCR-006-THRESHOLDS",
    "SCR-007-DESCRIPTION"
  ],
  "parameters": {
    "investigator_name": "Sarah Chen",
    "review_date": "2026-04-29",
    "viewport_width": 1456,
    "viewport_height": 816,
    "speaker_style": "Professional, concise, explanatory",
    "target_fp_rate": 30,
    "recall_floor": 95
  },
  "runtime_rules": {
    "preferred_locator_order": ["html id (#name)", "aria-label", "visible text", "fallback coordinates"],
    "wait_timeout_seconds": 20,
    "screenshot_policy": "capture_on_major_screen_change",
    "narration_policy": "speak_before_and_after_each_major_action",
    "do_not_infer_unreadable_content": true,
    "do_not_override_blocked_thresholds": true,
    "do_not_auto_clear_without_llm_explanation": true,
    "navigation_surface": "taskbar_dock_primary_desktop_icons_for_notes_calendar_only"
  },
  "key_selectors": {
    "desktop_icons": {
      "notes": "#desktop-notes-icon",
      "calendar": "#desktop-calendar-icon"
    },
    "dock": {
      "desktop": "#dock-desktop",
      "dashboard": "#dock-dashboard",
      "transactions": "#dock-transactions",
      "reports": "#dock-reports",
      "model": "#dock-model",
    }
  }
}
```

```

    "threshold": "#dock-threshold",
    "description": "#dock-description"
  },
  "dashboard_kpis": {
    "total": "#dashboard-kpi-total",
    "flagged": "#dashboard-kpi-flagged",
    "fraud": "#dashboard-kpi-fraud",
    "fprate": "#dashboard-kpi-fprate"
  },
  "tx_actions_template": {
    "fraud": "#tx-{tx_id}-fraud-btn",
    "clear": "#tx-{tx_id}-clear-btn",
    "escalate": "#tx-{tx_id}-escalate-btn"
  },
  "thresholds_template": {
    "row": "#threshold-row-{id}",
    "input": "#threshold-input-{id}",
    "approve": "#threshold-approve-btn-{id}",
    "revise": "#threshold-revise-btn-{id}",
    "admin_badge": "#threshold-admin-badge"
  },
  "live_activity": "#commentaryFeed",
  "menubar": "#menubar"
},
"completion_criteria": [
  "SCR-001-HOME opened and Live Activity verified",
  "SCR-002-DASHBOARD KPIs read and narrated",
  "SCR-002-DASHBOARD chart and risk donut explained",
  "SCR-003-TX-QUEUE opened and AI explanations reviewed",
  "Fraud confirmed for TXN-20240424-0064",
  "False positives cleared for TXN-20240424-0091 and TXN-20240424-0088",
  "SCR-004-REPORTS daily metrics narrated and PDF downloaded",
  "SCR-005-MODEL-PERF – all three model cards read",
  "SCR-005-MODEL-PERF – Audit Log reviewed",
  "SCR-006-THRESHOLDS – Admin Role validated",
  "SCR-006-THRESHOLDS – Crypto Exchange and Small International approvals submitted",
  "SCR-006-THRESHOLDS – Wallet Transfer Blocked status validated",
  "Final summary delivered"
]
}

```

21. Final Response Object Example

```
{
  "flow_id": "FLOW-FRAUDOS-INVEST-001",
  "flow_version": "1.0",
  "status": "completed",
  "screens_visited": [
    "SCR-001-HOME",
    "SCR-002-DASHBOARD",
    "SCR-003-TX-QUEUE",
    "SCR-004-REPORTS",
    "SCR-005-MODEL-PERF",
    "SCR-006-THRESHOLDS"
  ],
  "actions_completed": [
    {"action_id": "ACT-001", "screen_id": "SCR-001-HOME", "selector": "#commentaryFeed", "result": "home_screen_verified"},
    {"action_id": "ACT-002", "screen_id": "SCR-001-HOME", "selector": "#dock-dashboard", "result": "dashboard_opened"},
    {"action_id": "ACT-003", "screen_id": "SCR-002-DASHBOARD", "selector": "#dashboard-kpi-*", "result": "kpis_narrated"},
    {"action_id": "ACT-006", "screen_id": "SCR-001-HOME", "selector": "#dock-transactions", "result": "tx_queue_opened"},
    {"action_id": "ACT-008", "screen_id": "SCR-003-TX-QUEUE", "selector": "#tx-{id}-fraud-btn", "result": "fraud_confirmed_TXN-0064"},
    {"action_id": "ACT-009", "screen_id": "SCR-003-TX-QUEUE", "selector": "#tx-{id}-clear-btn", "result": "cleared_TXN-0091"},
    {"action_id": "ACT-010", "screen_id": "SCR-003-TX-QUEUE", "selector": "#tx-{id}-clear-btn", "result": "cleared_TXN-0088"},
    {"action_id": "ACT-011", "screen_id": "SCR-001-HOME", "selector": "#dock-reports", "result": "reports_opened"},
    {"action_id": "ACT-012", "screen_id": "SCR-004-REPORTS", "selector": "#reports-archive-table", "result": "compliance_report_narrated"},
    {"action_id": "ACT-013", "screen_id": "SCR-004-REPORTS", "selector": "#reports-today-download-btn", "result": "report_pdf_downloaded"},
    {"action_id": "ACT-014", "screen_id": "SCR-001-HOME", "selector": "#dock-model", "result": "model_perf_opened"},
    {"action_id": "ACT-015", "screen_id": "SCR-005-MODEL-PERF", "selector": "#model-perf-card-ensemble", "result": "model_performance_narrated"},
    {"action_id": "ACT-016", "screen_id": "SCR-005-MODEL-PERF", "selector": "#model-audit-table", "result": "audit_log_validated"},
    {"action_id": "ACT-017", "screen_id": "SCR-001-HOME", "selector": "#dock-threshold", "result": "thresholds_opened"},
    {"action_id": "ACT-018", "screen_id": "SCR-006-THRESHOLDS", "selector": "#threshold-admin-badge", "result": "admin_role_validated"},
    {"action_id": "ACT-019", "screen_id": "SCR-006-THRESHOLDS", "selector": "#threshold-approve-btn-{crypto}", "result": "threshold_approved_crypto_exchange"},
    {"action_id": "ACT-020", "screen_id": "SCR-006-THRESHOLDS", "selector": "#threshold-approve-btn-{small_intl}", "result": "threshold_approved_small_international"},
    {"action_id": "ACT-021", "screen_id": "SCR-006-THRESHOLDS", "selector": "#threshold-row-{wallet} .t-status.blocked", "result": "wallet_transfer_blocked_validated"}
  ],
  "humanoid_final_spoken_summary": "We reviewed all six FraudOS modules today. Navigation was performed through the taskbar dock (`#dock-*` icons), with the desktop showing only Notes and Calendar utility icons. On the Dashboard (SCR-002), 8 transactions were processed, all 8 flagged, and 5 confirmed as fraud, with FP Rate at 37.5% – above the 30% target from MET-01. In the Transaction Review Queue (SCR-003), I confirmed TXN-20240424-0064 as fraud ($34,200, Hong Kong, ensemble 0.97) and cleared two false positives. The Compliance Report (SCR-004) shows 94.2% review completion and 87.0% LLM helpful rate. On Model Performance (SCR-005), the Ensemble model leads with F1 0.955. On Thresholds (SCR-006), I submitted approvals for Crypto Exchange and Small International Transfer; Wallet Transfer Domestic was correctly blocked by the 95% recall floor. The new Description page (SCR-007) is available from the taskbar for read-only stakeholder review.",
  "recommended_next_action": "Monitor FP Rate over next 24 hours after approved threshold changes take effect through the BR-002 weekly cycle. If FP Rate drops below 30%, MET-01 target is met."
}
```

— End of FraudOS Virtual Humanoid User Flow Specification —